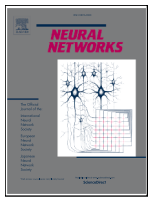
Neural Networks 206 (2027) 109865



Contents lists available at ScienceDirect

Neural Networks

journal homepage: www.elsevier.com/locate/neunet



Full Length Article



RE-DGCL: Reinforcement-evolutionary dynamic graph contrastive learning via proactive decisional evolution

Weidong Zhao [a,b], Jingjing Liang [a,b], Liqing Qiu [a,b,*]

[a] College of Computer Science and Engineering, Shandong University of Science and Technology, Qingdao, 266590, China
[b] Shandong Key Laboratory of Smart Mine Information Technology, Qingdao, 266590, China

ARTICLE INFO

Keywords:
Dynamic graph representation learning
Graph contrastive learning
Reinforcement learning
Decisional evolution

ABSTRACT

Dynamic graph representation learning is pivotal for modeling evolving complex systems. However, existing methods predominantly follow a "forced evolution" paradigm, in which node states are unconditionally updated upon every interaction. This passive strategy may introduce temporal redundancy and propagate high-frequency stochastic noise, leading to representational drift. Furthermore, an important yet underexplored challenge—termed the "Hard Evolutionary Boundary" problem—is identified in dynamic contrastive learning: conventional random negative sampling may fail to adequately distinguish between recently disconnected historical neighbors and irrelevant noise, resulting in a loose discriminative manifold. To address these limitations, this paper proposes RE-DGCL, a novel framework that reformulates dynamic graph modeling as a sequential decisional control task. RE-DGCL adopts a bifurcated architecture that explicitly decouples Stable Global Embeddings (SGE)—anchoring topological invariants via relational diffusion—from Adaptive Local Embeddings (ALE), which track volatile interaction pulses. To mitigate forced evolution, a Reinforcement Learning (RL)-driven Evolutionary Decision Action (EDA) mechanism is introduced, conceptualizing the node update process as an autonomous information bottleneck: the RL agent learns an "on-demand" gating policy that selectively filters out non-informative fluctuations. Simultaneously, an Evolutionary-aware Negative Sampling (ENS) module is designed to sharpen the contrastive boundary by identifying Hard Negative Constraints (HNC) from a historical proximity buffer. Extensive experiments on eight real-world datasets, together with random-edge perturbation tests, demonstrate that RE-DGCL achieves strong overall performance and exhibits improved robustness in sparse and noisy dynamic scenarios.

1. Introduction

Dynamic graphs serve as a powerful abstraction for modeling complex systems that evolve over time, such as social interaction networks, financial transaction streams, and biological signaling pathways (Alemany et al., 2023; Sharma et al., 2024). The core challenge of dynamic graph representation learning is to project these time-varying relational structures into a low-dimensional manifold that preserves both instantaneous local patterns and long-term structural invariants. These learned embeddings are essential for downstream predictive tasks, including dynamic link prediction, anomaly detection, and community evolution analysis (Huang et al., 2023; Qin et al., 2025; Yi et al., 2024).

Recent years have witnessed a shift from supervised methods to self-supervised paradigms, particularly Graph Contrastive Learning (GCL), to mitigate the scarcity of high-quality labels in dynamic environments (Chen et al., 2020; He et al., 2020). Foundational frameworks such as GRACE (Zhu et al., 2020) and GCA (Zhu et al., 2021) have demonstrated the effectiveness of maximizing mutual information between augmented topological views. To further capture temporal dynamics, methods such as TGN (Rossi et al., 2020) and DySAT (Sankar et al., 2020) have integrated memory modules and self-attention mechanisms to track evolving node states. Nevertheless, these methods predominantly follow a "forced evolution" paradigm, updating node embeddings at every observed interaction regardless of the semantic significance of the event. In real-world dynamic systems, many interactions represent stochastic noise or redundant fluctuations. This passive update strategy may introduce temporal redundancy and high-frequency noise, which obscures the underlying stable evolutionary trajectory of the network. Crucially, such indiscriminate updates give rise to what is termed "Representational Drift": the accumulation of non-informative stochastic pulses gradually erodes the structural anchors of node embeddings, resulting in a progressively degraded signal-to-noise ratio in dynamic representations.

* Corresponding author.
E-mail addresses: zwdslj@163.com (W. Zhao), 202483060138@sdust.edu.cn (J. Liang), qiuliqing2026@163.com (L. Qiu).

https://doi.org/10.1016/j.neunet.2026.109865
Received 30 April 2026; Received in revised form 28 August 2026; Accepted 1 November 2026
Available online 4 November 2026
0893-6080/© 2026 Elsevier Ltd. All rights are reserved, including those for text and data mining, AI training, and similar technologies.

The discriminative power of contrastive learning is also strongly influenced by the quality of negative samples. Most existing dynamic GCL frameworks rely on random uniform sampling to construct the negative semantic space. Such a strategy predominantly selects "easy negatives"—nodes that are topologically remote and temporally uncorrelated—providing insufficient contrastive signals and yielding a loose decision boundary that may inadequately characterize the fine-grained evolutionary manifold. In particular, existing methods struggle to distinguish nodes that were previously relevant but are currently inactive from those that have never interacted. This challenge is defined as the "Hard Evolutionary Boundary" (HEB) problem. Without explicitly penalizing recently-disconnected neighbors, the model may suffer from "Manifold Collapse", wherein the distinction between historical dependency and current interaction becomes blurred in the latent space.

To address these limitations, RE-DGCL (Reinforcement-Evolutionary Dynamic Graph Contrastive Learning) is proposed—a novel self-supervised framework that transitions dynamic graph modeling from passive aggregation to proactive decisional evolution. Unlike conventional approaches, RE-DGCL conceptualizes the node update process as an autonomous decision-making task controlled by an RL agent. An Evolutionary Decision Action (EDA) mechanism is introduced, driven by a learnable RL-based gating policy. For each incoming interaction pulse, the agent evaluates the necessity of state transition, selectively updating the Adaptive Local Embedding (ALE) to filter out temporal noise. To sharpen the contrastive boundary, an Evolutionary-aware Negative Sampling (ENS) module is further designed. By leveraging a historical interaction buffer and a curriculum pacing function, ENS identifies Hard Negative Constraints (HNC)—nodes with high temporal proximity but no current connection—thereby enforcing a rigorous discriminative boundary. Finally, a hierarchical objective aligns the ALE with Stable Global Embeddings (SGE) via a closed-loop reward signal, ensuring multi-scale representational consistency.

Despite substantial progress in temporal and structural representation learning, existing methods mainly focus on how node states evolve after an interaction is observed, while paying less attention to whether every interaction should trigger state evolution. RE-DGCL therefore formulates representation updating as a proactive sequential decision problem, allowing node states to be selectively preserved or evolved according to their structural and temporal context.

The main contributions are summarized as follows:

1. A proactive evolutionary sensing strategy is proposed, which constructs a multi-granular observation space and identifies Hard Negative Constraints (HNC) to define rigorous evolutionary boundaries for contrastive learning.
2. An RL-driven Adaptive Decisional Encoding architecture is introduced. Through "on-demand" state updates governed by a learnable gating policy, the model autonomously suppresses temporal redundancy and stochastic interaction noise.
3. A counterfactual reward-driven optimization framework is designed, in which both feasible evolutionary transitions are evaluated under a shared structural-fidelity criterion to determine the relative utility of the selected action, thereby enabling self-correction of the decision policy and forming a closed-loop mechanism for high-fidelity representation learning.
4. Extensive experiments on eight real-world datasets and additional random-edge perturbation tests demonstrate the effectiveness and robustness of RE-DGCL across heterogeneous dynamic scenarios.

2. Related work

2.1. Dynamic graph representation learning

Dynamic graph representation learning aims to produce expressive low-dimensional node embeddings that simultaneously capture instantaneous structural states and long-term temporal evolution. Existing methods fall into two broad categories: discrete-time models and continuous-time models (Kazemi et al., 2020; Skarding et al., 2021).

Discrete-time models represent a dynamic graph as an ordered sequence of static snapshots, applying GNN-based spatial encoders within each snapshot before aggregating temporal signals across time. GCN (Kipf & Welling, 2016) performs neighborhood aggregation via the normalized graph Laplacian, while GraphSAGE (Hamilton et al., 2017) extends this paradigm inductively through learnable neighbor sampling and aggregation. To explicitly model inter-snapshot transitions, a major line of research couples spatial GNN encoders with recurrent sequence models. EvolveGCN (Pareja et al., 2020) evolves the GCN weight matrices themselves through a recurrent network, allowing the model's inductive bias to adapt to structural drift without explicitly storing node-level states. GCRN (Seo et al., 2018) and STGCN (Yu et al., 2018) pursue a parallel design in which GCN layers capture spatial dependencies while RNN or convolutional modules model temporal transitions, achieving strong results in spatiotemporal forecasting. DynGEM (Goyal et al., 2018) learns per-snapshot embeddings through autoencoders while regularizing temporal smoothness by penalizing large embedding shifts between adjacent snapshots.

Continuous-time models directly process raw interaction event streams in chronological order, preserving full temporal resolution. CTDNE (Nguyen et al., 2018) extends the random-walk paradigm to continuous time by enforcing temporal ordering constraints along walk paths. DyGNN (Ma et al., 2020) employs LSTM-like memory cells to update both the interacting nodes and all their neighborhoods upon each interaction event. JODIE (Kumar et al., 2019a) and DyRep (Trivedi et al., 2019) are landmark contributions that use coupled RNNs to jointly update the embeddings of interacting node pairs, introducing the concept of a continuously evolving embedding trajectory. TGAT (Xu et al., 2020) integrates temporal and topological neighborhood features through a modified self-attention mechanism in which functional time encoding replaces conventional positional encoding. TGN (Rossi et al., 2020) combines an explicit per-node memory module with a temporal graph attention mechanism for neighborhood aggregation, achieving superior performance in both transductive and inductive settings. DySAT (Sankar et al., 2020) captures spatio-temporal dynamics through structural and temporal self-attention applied separately within and across snapshots.

Despite their advances, both families predominantly follow a passive "forced evolution" paradigm, obligatorily updating node embeddings upon every observed interaction regardless of semantic significance. In real-world dynamic systems, many interactions represent stochastic noise or redundant fluctuations. Blindly propagating such noisy information can contaminate node representations, leading to temporal redundancy or even model collapse (Li et al., 2024; Rossi et al., 2020).

Recently, state-space models have been introduced into dynamic graph learning to improve long-range temporal modeling efficiency. DyG-Mamba (Li et al., 2025b) integrates state-space modeling with temporal graph learning to capture long-term dependencies with reduced computational complexity. However, such approaches primarily focus on how temporal information is propagated, whereas RE-DGCL addresses the complementary question of whether each observed interaction should trigger state evolution.

2.2. Negative sampling in graph learning

Negative sampling was originally introduced to accelerate training of the Skip-Gram model in Word2Vec (Mikolov et al., 2013), and was subsequently adopted in static network embedding methods such as DeepWalk (Perozzi et al., 2014), Node2vec (Grover & Leskovec, 2016), and LINE (Tang et al., 2015). These foundational methods typically employ random uniform sampling based on node degree distributions to counterbalance the limited availability of positive edges. As representation learning evolved, researchers in recommendation systems recognized the limitations of random sampling and developed hard nega-

tive sampling techniques—such as PinSAGE (Ying et al., 2018)—which utilize random walks to surface topologically relevant but not directly connected nodes, thereby providing stronger contrastive gradients and higher-resolution decision boundaries.

Applying standard random negative sampling directly to dynamic graphs, however, presents significant challenges. Dynamic graphs are characterized by highly skewed temporal distributions and evolving topologies. Recent studies (Gao et al., 2024) have highlighted two pervasive traits in dynamic networks: historical dependency (nodes frequently reconnect with past neighbors) and temporal proximity preference (nodes preferentially interact with recently connected neighbors). Standard uniform sampling entirely ignores this temporal dimension, predominantly selecting "easy negatives"—nodes that are topologically remote and temporally uncorrelated. Consequently, Dynamic Graph Neural Networks (DGNNs) trained with such easy negatives tend to learn loose discriminative boundaries and suffer from the "Hard Evolutionary Boundary" problem, failing to distinguish nodes with frequent historical interactions from those that have never interacted.

To mitigate this issue, more sophisticated strategies, including dynamic negative sampling (Zhang et al., 2013) and GAN-based adversarial sampling (Bose et al., 2018), have been proposed. Yet abruptly introducing highly informative hard negatives early in training often destabilizes convergence in GCL. Inspired by cognitive science, Curriculum Learning (Bengio et al., 2009) has been applied to graph representation learning (Wei et al., 2023) to organize training samples in an easy-to-difficult sequence, allowing the model to establish stable structural patterns before refining complex evolutionary boundaries.

Recent studies have further examined false negatives and implicit conflicts in graph contrastive learning. ProGCL (Xia et al., 2022) introduces probabilistic hard-negative estimation to reduce false-negative interference, while subsequent approaches improve negative selection by resolving implicit conflicts (He et al., 2024) or enhancing sampling efficiency (Huang et al., 2025). Unlike these methods, which mainly assess negative quality from the learned representation space, ENS defines evolutionary hardness according to historical interaction proximity and current disconnection, specifically targeting dynamic graph evolution.

Driven by these insights, RE-DGCL departs from naive uniform sampling by proposing an Evolutionary-aware Negative Sampling (ENS) module. ENS leverages a node-specific History Interaction Buffer to systematically retrieve "just-disconnected" nodes as Hard Negative Constraints (HNC), and integrates a curriculum pacing function to dynamically regulate the mixture of random noise and HNCs during training. This ensures a stable progression from learning broad topological features to capturing fine-grained evolutionary manifolds, thereby maximizing the discriminative power of the contrastive objective.

2.3. Reinforcement learning in graph learning

Reinforcement Learning (RL) is well suited to discrete, non-differentiable sequential decision-making problems by learning a policy that maximizes expected cumulative rewards through environment interaction (Kaelbling et al., 1996). In the realm of static GNNs, RL has been primarily applied to optimize graph architectures (e.g., GraphNAS (Gao et al., 2020)) or to design neighbor sampling mechanisms that constrain receptive field expansion (Li et al., 2016; You et al., 2019).

In the context of dynamic graphs, the integration of RL is strongly motivated by the need for robustness against transient noise. State-of-the-art continuous-time dynamic graph models, such as TGN (Rossi et al., 2020), JODIE (Kumar et al., 2019a), and DyRep (Trivedi et al., 2019), predominantly follow a "forced evolution" paradigm: the embeddings of interacting nodes (and often their neighborhoods) are obligatorily updated upon every observed interaction. While this design captures high-frequency temporal dynamics, it introduces a critical vulnerability—in real-world scenarios, newly established links frequently carry stochastic noise, accidental interactions, or outdated information (Li et al., 2024). Blindly updating and propagating such noisy knowledge can severely contaminate node representations, leading to temporal redundancy or model collapse.

Recent pioneering works have begun to address this by conceptualizing the node update process as a sequential decision problem. AdaDyGNN (Li et al., 2024), for example, employs an RL agent to adaptively select which neighbor nodes should be updated upon a new interaction, using the stability of local graph structures as a reward signal, thereby preventing the propagation of noisy interactions.

Beyond dynamic graph updating, recent studies have also explored proactive decision-making systems based on structural information (Zeng et al., 2026). Such approaches demonstrate that graph structures can provide meaningful signals for making early decisions rather than merely reacting to observed changes. Moreover, hierarchical decision-making frameworks based on structural principles (Zeng et al., 2025) further reveal that complex graph systems can benefit from multi-level decision processes, where global structural patterns guide local adaptive actions. These perspectives provide valuable insights for designing intelligent decision mechanisms in graph learning, where structural information serves as an important basis for adaptive decision-making.

RE-DGCL significantly advances this trajectory by formulating the core temporal encoding process as an autonomous, closed-loop RL task. Unlike prior methods that merely select which neighbors to update, the Adaptive Decisional Encoding stage in RE-DGCL introduces an RL-driven Decisional Gating policy. Modeled as a Bernoulli distribution, this policy autonomously dictates the micro-level state transitions of the ALE—choosing between an "Evolutionary Update" and "State Maintenance" for each incoming message pulse. By optimizing this policy via the REINFORCE algorithm guided by the counterfactual Evolutionary Fidelity Reward (EFR), RE-DGCL transforms passive, noise-susceptible aggregation into proactive, "on-demand" evolution. This design effectively creates an information bottleneck that suppresses high-frequency stochastic noise while preserving semantically significant evolutionary trajectories.

2.4. Graph contrastive learning

Graph Contrastive Learning (GCL) has emerged as a dominant paradigm for self-supervised graph representation learning, operating by maximizing agreement between augmented views of the same entity while discriminating against representations of other entities (Chen et al., 2020; He et al., 2020). The effectiveness of a GCL framework depends critically on two core components: the data augmentation strategy and the contrastive objective.

Early foundational work established a local-global mutual information maximization paradigm. DGI (Veličković et al., 2018) contrasts patch-level node representations against a global graph summary, driving the encoder to capture globally informative structure. However, its view generation relies on simple feature permutation with limited structural diversity. GRACE (Zhu et al., 2020) introduced a node-level contrastive paradigm by generating two views through independent random augmentations—node feature masking and edge dropping—and treating the same node across views as a positive pair. GCA (Zhu et al., 2021) improved upon this by introducing centrality-guided adaptive augmentation that selectively preserves topologically critical structures while perturbing peripheral elements. MVGRL (Hassani & Khasahmadi, 2020) demonstrated the value of multi-scale contrast by pairing a raw topological view with a graph-diffusion-smoothed view, capturing both local and global neighborhood signals within a unified contrastive framework.

A persistent challenge in instance-level contrastive methods is the false negative problem, where semantically similar nodes are inadvertently treated as negatives. Prototypical approaches mitigate this by contrasting node representations against cluster centroids rather than individual instances (Li et al., 2021). BGRL (Thakoor et al., 2022) takes an orthogonal route, eliminating negative sampling entirely by adapting the BYOL framework (Grill et al., 2020) to graphs through an asymmet-

Table 1
Key mathematical notations of RE-DGCL.

Notation	Description
G	A sequence of dynamic graph snapshots $\{G_1, G_2, \ldots, G_T\}$
G_t	Graph snapshot at timestamp t
V, E_t	Fixed node set and time-varying edge set
A_t, X_t	Adjacency matrix and node feature matrix at time t
$\mathcal{V}_{\text{EVS}}$	Evolutionary View Set: $\{G^{\text{fev}}, G^{\text{rev}}, G^{\text{mev}}\}$
$\mathcal{B}_i$	History Interaction Buffer for node i
$last_ts(i, v)$	Most recent interaction timestamp between node i and v
H_{SGE}	Stable Global Embedding capturing structural invariants
Z_{ALE}	Adaptive Local Embedding capturing temporal evolution
$m_{i,t}$	Evolutionary message pulse synthesized from local neighbors
$a_{i,t} \in \{0, 1\}$	Evolutionary Decision Action (1: Update, 0: Maintain)
π_θ	Decisional gating policy parameterized by θ
$R_{i,t}$	Evolutionary Fidelity Reward for policy feedback
$\mathcal{L}_{\text{con}}, \mathcal{L}_{\text{RL}}$	Hierarchical consistency loss and RL-driven policy loss

ric online-target encoder architecture, learning purely from positive pair consistency without risk of false-negative contamination.

More recent studies have extended graph contrastive learning beyond instance-level consistency by incorporating higher-level structural semantics. Community-invariant graph contrastive learning (Tan et al., 2024) introduces community-aware constraints to preserve stable semantic structures across augmented graph views, improving robustness against structural perturbations. Different from instance discrimination objectives, these methods emphasize maintaining community-level invariance to enhance representation reliability.

For dynamic graphs, applying static GCL methods directly is insufficient, as augmentations must respect temporal structure. CPDG (Bei et al., 2023) proposes temporal-neighborhood-based contrastive pre-training for dynamic GNNs. CLDG++ (Xu et al., 2024b) generates multi-scale temporal views via Personalized PageRank and heat kernels, contrasting them to jointly capture local interaction dynamics and global structural correlations across timescales. In addition to contrastive representation learning, dynamic graph clustering methods have investigated how to preserve evolving structural patterns over time. Recent work on revisiting dynamic graph clustering via matrix factorization (Li et al., 2025a) highlights the importance of maintaining persistent community structures during graph evolution. Although these clustering-oriented approaches are not designed for dynamic link prediction, they further demonstrate the necessity of distinguishing stable structural information from transient interactions, which aligns with the motivation of separating global stability and local adaptation in RE-DGCL. These efforts, however, largely treat temporal modeling and contrastive learning as independent concerns, without a unified mechanism that leverages global structural stability to calibrate local temporal representations against noise—a gap that RE-DGCL directly addresses.

3. Preliminaries

This section establishes the formal definitions and notations essential for understanding the RE-DGCL framework. The primary mathematical notations are summarized in Table 1.

3.1. Basic definitions and data structures

Definition 3.1 (Dynamic Graph)**.** A dynamic graph is formally represented as a sequence $G = \{G_1, G_2, \ldots, G_T\}$, capturing the temporal evolution of topology and node attributes across T discrete snapshots.

Definition 3.2 (Evolutionary View Set (EVS))**.** To model multi-scale graph dynamics, an augmentation space $\mathcal{V}_{\text{EVS}} = \{G^v \mid v \in \mathcal{T}\}$ is defined as a collection of semantic manifolds derived from the raw snapshot G_t. The EVS provides a multi-granular data foundation, enabling the model to jointly capture instantaneous interactions, structural resilience, and long-term evolutionary trends.

Definition 3.3 (History Interaction Buffer)**.** For each node $i \in V$, a memory-centric repository $\mathcal{B}_i$ archives the historical interaction trajectory of its local neighborhood. This buffer serves as the structural support for identifying Hard Negative Constraints (HNC), which define the most semantically challenging boundaries in the dynamic latent space.

3.2. Decisional representation principles

Definition 3.4 (Decisional State Transition)**.** Node representation is framed as a bifurcated system comprising a Stable Global Embedding (H_{SGE}) and an Adaptive Local Embedding (Z_{ALE}). The evolution of Z_{ALE} is formulated as a discrete decision-making problem, where state transitions are controlled by a gating signal $a_{i,t}$, enabling "evolution-on-demand" and suppression of temporal noise.

3.3. Problem formulation

Problem 3.1 (Dynamic Link Prediction)**.** Given a sequence G, the objective is to learn a reward-driven decisional policy and an embedding function that improve predictive accuracy for future edge sets. This is achieved by aligning representations across evolutionary scales while ensuring representational fidelity through reward-driven optimization.

4. Method

The RE-DGCL (Reinforcement-Evolutionary Dynamic Graph Contrastive Learning) framework is conceptualized as an autonomous evolutionary agent that transcends the limitations of passive feature extraction by integrating proactive environment sensing, intelligent state gating, and closed-loop reward-driven optimization. As shown in Fig. 1, the overall framework of RE-DGCL consists of three core modules: proactive evolutionary sensing, adaptive decisional encoding and reward-driven optimization. The following subsections provide an in-depth analysis of these three key components.

4.1. Stage 1: Proactive evolutionary sensing

The objective of this stage is to construct a diverse Evolutionary View Set (EVS) and define a rigorous Hard Negative Constraints (HNC) space, thereby providing a robust learning environment for subsequent stages.

4.1.1. Evolutionary view set (EVS) generation

Given a raw dynamic graph snapshot $G_t = (V, E_t)$ at time t with adjacency matrix $A_t \in \{0, 1\}^{N \times N}$ and node feature matrix $X_t \in \mathbb{R}^{N \times F}$, transformation functions $\mathcal{T}(\cdot)$ are applied to generate the EVS, denoted as $\mathcal{V}_{\text{EVS}} = \{G^{\text{fev}}, G^{\text{rev}}, G^{\text{mev}}\}$.

- **FEV (Factual Evolutionary View):** The factual view G^{fev} is equivalent to the raw snapshot G_t, capturing high-frequency instantaneous interaction patterns without any perturbation.
- **REV (Resilient Evolutionary View):** The resilient view G^{rev} is generated by applying an edge-dropout mask $M_t \in \{0, 1\}^{N \times N}$, where each entry $M_{ij} \sim \text{Bernoulli}(1 - p_{\text{drop}})$. The resulting adjacency matrix is $A_t^{\text{rev}} = A_t \odot M_t$, where p_{drop} is the dropout rate simulating link uncertainty.
- **MEV (Macro-Evolutionary View):** The macro view G^{mev} captures low-frequency structural trends by aggregating historical snapshots within a sliding window of size W. Its adjacency matrix is defined as $A_t^{\text{mev}} = \bigvee_{k=t-W+1}^{t} A_k$, where $\bigvee$ denotes the logical OR operator.

4.1.2. Evolutionary-aware negative sampling (ENS)

The efficacy of contrastive learning is strongly influenced by the informativeness of negative pairs. In dynamic graph scenarios, standard random uniform sampling predominantly selects "easy negatives"—nodes that are topologically remote and temporally uncorrelated—resulting in a loose discriminative boundary and a suboptimal representation of the evolutionary manifold. To address this

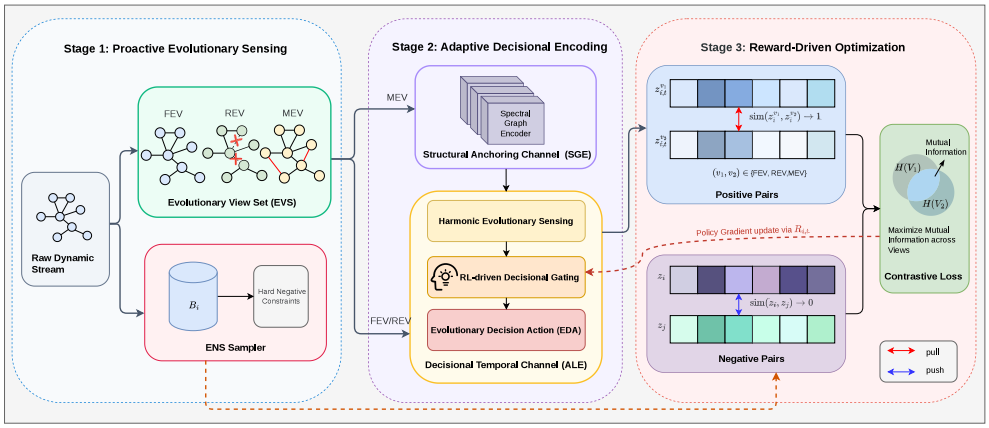


Fig. 1. The overall framework of RE-DGCL.

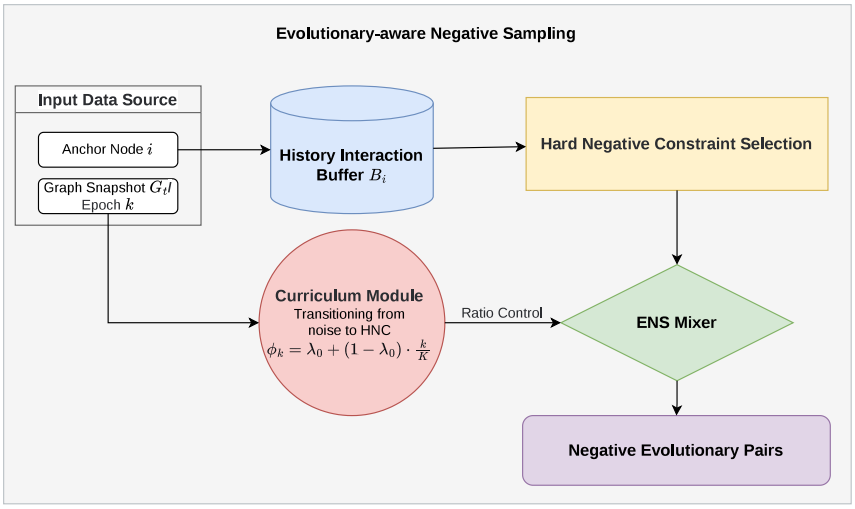


Fig. 2. The internal workflow of evolutionary-aware negative sampling.

limitation, the Evolutionary-aware Negative Sampling (ENS) module is proposed to systematically construct Hard Negative Constraints (HNC). As illustrated in Fig. 2, the ENS module operates through a two-stage pipeline: first, it retrieves historically relevant yet currently disconnected nodes from a memory buffer; second, it employs a curriculum pacing strategy to progressively integrate these hard negatives into the contrastive learning process, thereby sharpening the evolutionary decision boundary.

A. Definition of Evolutionary Hardness. The "hardness" of a negative sample is defined not merely by feature similarity, but by its temporal proximity to the active manifold. Specifically, the most informative negatives are nodes that previously maintained frequent interactions with the anchor node but are currently in a disconnected state—pairs that represent the "critical boundaries" of network evolution.

B. Memory-driven Sampling Mechanism. To retrieve such informative constraints, a node-specific History Interaction Buffer $\mathcal{B}_i$ is maintained for each node $i \in V$. The buffer archives the set of historical neighbors and their corresponding final interaction timestamps. Formally, for a given anchor node i at snapshot G_t, the ENS module identifies the hard negative v_{neg} by solving the following proximity optimization:

$$v_{\text{neg}} = \arg \min_{v \in \mathcal{B}_i, (i,v) \notin E_t} \left| t_{\text{curr}} - \text{last_ts}(i, v) \right| \tag{1}$$

where t_{curr} denotes the current temporal coordinate and $\text{last_ts}(i, v)$ is the most recent timestamp of interaction between node i and v. By minimizing this temporal latency, ENS selects "just-disconnected" nodes that reside in the high-density region of the node's potential interaction space, thereby providing the strongest contrastive signal to the ALE.

It should be noted that historical disconnection does not necessarily imply permanent semantic irrelevance. A historically connected node may reconnect with the anchor node in subsequent snapshots and thus constitute a potential false negative from a long-term perspective. Therefore, HNCs in RE-DGCL are interpreted as snapshot-conditional evolutionary constraints rather than permanently irrelevant node pairs. Specifically, ENS distinguishes nodes according to their current connectivity state while retaining their historical proximity as an indicator of contrastive difficulty.

C. Curriculum Pacing for Decisional Stability. Introducing high-difficulty HNC at the early stages of training may destabilize the EDA policy and lead to suboptimal convergence. Therefore, a Curriculum Pacing Function ϕ_k is incorporated to progressively transition the negative pool from stochastic noise to evolutionary constraints:

$$\phi_k = \lambda_0 + (1 - \lambda_0)\frac{k}{K_{\text{cur}}}, \tag{2}$$

The coefficient ϕ_k regulates the mixture ratio of HNC and random negatives at epoch k. This hierarchical curriculum ensures that the RL agent first establishes a stable structural baseline before refining the decisional manifold against nuanced evolutionary boundaries. Meanwhile, gradually increasing the exposure to HNCs prevents potentially ambiguous historical pairs from dominating the optimization at the early train-

ing stage, thereby mitigating, although not completely eliminating, the adverse effect of possible false negatives.

4.2. Stage 2: Adaptive decisional encoding

The Adaptive Decisional Encoding stage serves as the cognitive core of RE-DGCL. Unlike existing dynamic graph representation methods that suffer from temporal redundancy and over-smoothing, this stage implements a bifurcated architecture that explicitly decouples *topological invariants* from *volatile evolutionary pulses*. By orchestrating a Structural Anchoring Channel and a Decisional Temporal Channel, the framework transforms passive snapshot aggregation into an active, decision-driven state management process.

4.2.1. Structural anchoring channel: SGE extraction

A key challenge in dynamic networks is the spatial sparsity and transient noise present in discrete snapshots. To provide a reliable structural reference for the evolving system, the Structural Anchoring Channel extracts Stable Global Embeddings (SGE), denoted as $H_{\text{SGE}} \in \mathbb{R}^{N\times d}$.

Global community motifs are modeled by smoothing spatial irregularities through a K-step iterative relational diffusion process. Given the initial feature projection $H^{(0)}_{\text{SGE}} = X_t W_{\text{proj}}$, the propagation at step k is defined as:

$$H^{(k)}_{\text{SGE}} = (1-\beta)H^{(0)}_{\text{SGE}} + \beta \tilde{D}_t^{-1/2} A_t \tilde{D}_t^{-1/2} H^{(k-1)}_{\text{SGE}} \tag{3}$$

where $\beta \in (0,1)$ is the diffusion coefficient governing the trade-off between local feature preservation and multi-hop neighborhood context, and $\tilde{D}_t^{-1/2} A_t \tilde{D}_t^{-1/2}$ represents the symmetrically normalized adjacency matrix.

By transcending first-order connectivity, the SGE captures the latent community structure that remains relatively invariant within a temporal neighborhood. This topological stability acts as a *semantic anchor*, filtering out transient link jitter and providing a global coordinate system for calibrating subsequent local updates.

4.2.2. Decisional temporal channel: ALE evolution

Parallel to global anchoring, the Decisional Temporal Channel governs micro-level state transitions by generating Adaptive Local Embeddings (ALE), denoted as Z_{ALE}. The node state update is formulated not as deterministic accumulation, but as a discrete Markovian decision problem controlled by an RL agent.

A. Harmonic Evolutionary Sensing. To capture the periodic rhythms and bursty nature of interactions, temporal latencies $\Delta t_{ij} = t_{\text{curr}} - t_{\text{edge}}$ are projected into a high-dimensional harmonic manifold $\Phi(\Delta t_{ij}) \in \mathbb{R}^{d_{\text{time}}}$ using a set of learnable frequency bases. The incoming evolutionary message pulse $m_{i,t}$ is synthesized via a time-aware attention mechanism:

$$m_{i,t} = \sum_{j\in\mathcal{N}_i} \alpha_{ij,t}\left(Z^{\text{ALE}}_{j,t-1} + \Phi(\Delta t_{ij})\right) \tag{4}$$

where $\alpha_{ij,t}$ is the attention coefficient computed by measuring the compatibility between the anchor node's history $Z^{\text{ALE}}_{i,t-1}$ and the incoming neighbor stimuli. This ensures that the message pulse selectively emphasizes temporally relevant neighbors.

B. RL-driven Decisional Gating Policy. Standard Recurrent Neural Networks (RNNs) in dynamic graphs often overfit to high-frequency noise by forcing an update at every interaction. To alleviate this temporal redundancy, an RL-based gating policy π_θ is introduced. For each node i, the agent perceives a comprehensive observation state $s_{i,t} = [Z^{\text{ALE}}_{i,t-1} \| m_{i,t}]$, representing the fusion of long-term memory and the current interaction impulse. As illustrated in Fig. 3, the policy network processes this observation state through a series of transformations to produce a gating decision, determining whether the node should evolve or maintain its current state. Formally, the policy network π_θ maps $s_{i,t}$ to a Bernoulli distribution, from which the Evolutionary Decision Action (EDA) $a_{i,t} \in \{0,1\}$ is sampled:

$$\pi_\theta(a_{i,t}|s_{i,t}) = \text{Sigmoid}(W_\pi \cdot \sigma(W_s s_{i,t} + b_s) + b_\pi) \tag{5}$$

$$a_{i,t} \sim \text{Bernoulli}(\pi_\theta(s_{i,t})) \tag{6}$$

The action $a_{i,t} = 1$ signifies an *Evolutionary Update*, while $a_{i,t} = 0$ represents *State Maintenance*. This mechanism effectively functions as a stochastic information bottleneck, permitting only the most semantically significant pulses to alter the node's trajectory.

C. Decisional State Transition. Following the sampled EDA, the final evolution of the ALE is executed through a gated transition. This on-demand evolution mechanism ensures that the node state remains robust against stochastic jitter:

$$Z^{\text{ALE}}_{i,t} = (1-a_{i,t}) \cdot Z^{\text{ALE}}_{i,t-1} + a_{i,t} \cdot \text{GRU}(m_{i,t}, Z^{\text{ALE}}_{i,t-1}) \tag{7}$$

By optimizing the policy π_θ via evolutionary rewards detailed in Section 4.3, the model learns to suppress non-informative interactions. The resulting ALE thus provides a sparse yet high-fidelity representation of the micro-interaction trajectory, calibrated by the structural anchors of the SGE.

4.3. Stage 3: Reward-driven optimization

The final stage of RE-DGCL implements a joint optimization framework that synergistically refines node representations and the decisional gating policy. Unlike conventional dynamic GNNs that optimize solely

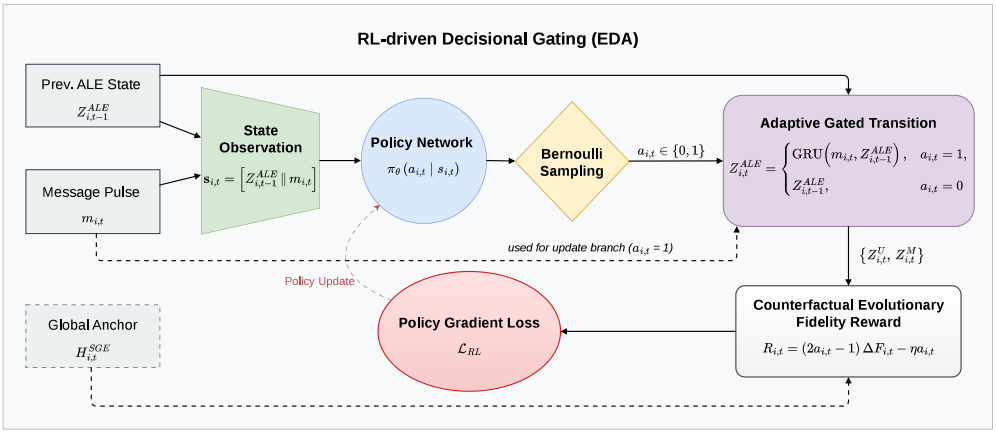


Fig. 3. The internal workflow of RL-driven decisional gating.

for reconstruction or link prediction accuracy, a reward-driven closed-loop mechanism is introduced. This stage aims to maximize mutual information across multiple scales while ensuring that the RL agent learns an effective "evolution-on-demand" strategy.

4.3.1. Cross-scale consistency (CSC) optimization

To calibrate the volatile ALE using the stable SGE as a structural reference, the Cross-Scale Consistency (CSC) objective is proposed. In each evolutionary view $v \in \mathcal{V}_{\text{EVS}}$, SGE provides a noise-resistant community-level context. Aligning the ALE with the corresponding SGE ensures that micro-level updates do not deviate from the underlying macro-structural trajectory.

For a given view v, the CSC loss is formulated using the Noise Contrastive Estimation (NCE) framework:

$$\mathcal{L}_{\text{csc}}(v) = -\frac{1}{N}\sum_{i=1}^{N}\log\frac{\exp(\text{sim}(Z_{i,v}^{\text{ALE}}, H_{i,v}^{\text{SGE}})/\tau)}{\sum_{j=1}^{N}\exp(\text{sim}(Z_{i,v}^{\text{ALE}}, H_{j,v}^{\text{SGE}})/\tau)} \tag{8}$$

where $\text{sim}(\cdot)$ denotes cosine similarity, τ is a temperature hyperparameter controlling the concentration of the distribution, and N is the number of nodes in a mini-batch.

Minimizing $\mathcal{L}_{\text{csc}}$ effectively acts as a *semantic regularizer*, forcing the Decisional Temporal Channel to prioritize features consistent with the long-term community motifs captured by the Structural Anchoring Channel.

4.3.2. Hierarchical objective with hard negative penalty

To further enhance the discriminative power of RE-DGCL, the CSC loss is integrated into a hierarchical objective function that simultaneously manages inter-view consistency and enforces the Hard Negative Constraints (HNC) retrieved in Stage 1:

$$\mathcal{L}_{\text{con}} = \sum_{v\in\mathcal{V}_{\text{EVS}}}\mathcal{L}_{\text{csc}}(v) + \delta\mathcal{L}_{\text{erc}}(\mathcal{V}_{\text{EVS}}) + \gamma\mathcal{L}_{\text{HNC}} \tag{9}$$

where $\mathcal{L}_{\text{erc}}$ ensures that representations of the same node are invariant across different augmented views (FEV, REV, and MEV). The HNC penalty $\mathcal{L}_{\text{HNC}}$ is defined to explicitly push away "recently disconnected" historical neighbors:

$$\mathcal{L}_{\text{HNC}} = -\sum_{(i,j)\in\mathcal{P}_{\text{hnc}}}\log(1-\sigma(Z_{i,t}^{\text{ALET}}Z_{j,t}^{\text{ALE}})) \tag{10}$$

where $\mathcal{P}_{\text{hnc}}$ denotes the set of HNC pairs. Here, δ and γ are balancing coefficients. By penalizing the similarity of HNC pairs, the model learns to identify fine-grained evolutionary boundaries, distinguishing between latent semantic similarity and temporal interaction decay.

4.3.3. Policy refinement via counterfactual EFR

The RL agent's policy π_θ is optimized through the Evolutionary Fidelity Reward (EFR), which evaluates whether an evolutionary state transition provides a meaningful structural benefit. A reward computed solely from the state produced after the selected action may introduce an action-dependent evaluation bias, since the updated ALE representation is directly generated from the current evolutionary message. To avoid such one-sided post-action evaluation, RE-DGCL adopts a counterfactual reward mechanism that evaluates both feasible state transitions under an identical action-independent criterion before measuring the relative utility of the selected action.

For node i at snapshot t, two counterfactual candidate states are first constructed. The evolutionary-update candidate is obtained by integrating the current evolutionary message $m_{i,t}$ with the previous ALE state:

$$Z_{i,t}^{U} = \text{GRU}\left(m_{i,t}, Z_{i,t-1}^{\text{ALE}}\right), \tag{11}$$

whereas the state-maintenance candidate directly preserves the previous representation:

$$Z_{i,t}^{M} = Z_{i,t-1}^{\text{ALE}}. \tag{12}$$

Rather than evaluating only the actually selected state, both candidates are assessed using the same action-independent structural-fidelity criterion. Specifically, the corresponding SGE $H_{i,t}^{\text{SGE}}$, generated by the structurally stable encoding channel, serves as a structural anchor:

$$F_{i,t}^{\text{upd}} = \text{sim}\left(Z_{i,t}^{U}, H_{i,t}^{\text{SGE}}\right), \quad F_{i,t}^{\text{keep}} = \text{sim}\left(Z_{i,t}^{M}, H_{i,t}^{\text{SGE}}\right). \tag{13}$$

The relative evolutionary fidelity gain is then defined as

$$\Delta F_{i,t} = F_{i,t}^{\text{upd}} - F_{i,t}^{\text{keep}}. \tag{14}$$

Based on this counterfactual comparison, the EFR is formulated as

$$R_{i,t} = (2a_{i,t}-1)\Delta F_{i,t} - \eta a_{i,t}. \tag{15}$$

where $\eta \geq 0$ is the transition cost used to discourage unnecessary state changes. The reward becomes $R_{i,t}(a_{i,t}=1) = \Delta F_{i,t} - \eta$ for an update and $R_{i,t}(a_{i,t}=0) = -\Delta F_{i,t}$ for state maintenance. Hence, updating is preferred when $2\Delta F_{i,t} > \eta$.

Because both candidates are evaluated independently of the sampled action using the same SGE-based criterion, the reward avoids directly comparing an action-generated state with the message used to generate it. This removes the one-sided self-feedback bias and allows the policy to compare the relative utility of updating and maintaining the state.

The policy parameters θ are optimized by minimizing the following REINFORCE policy-gradient loss:

$$\mathcal{L}_{\text{RL}} = -\mathbb{E}_{\pi_\theta}\left[\sum_t\sum_i \log\pi_\theta(a_{i,t} \mid s_{i,t})R_{i,t}\right]. \tag{16}$$

Accordingly, the counterfactual reward encourages EDA to favor state evolution when the candidate update provides sufficient structural-fidelity gain. The relatively stable SGE therefore guides local ALE transitions, yielding a structurally grounded hierarchical decision process.

More specifically, the discrete gating policy is expected to be most beneficial when incoming interactions exhibit heterogeneous evolutionary utility. According to the reward formulation above, an update is favored when $2\Delta F_{i,t} > \eta$, indicating that the structural-fidelity gain is sufficiently large to offset the transition cost; otherwise, state maintenance is preferred. Since both counterfactual candidates are evaluated against the same action-independent SGE anchor before the selected action is rewarded, their relative comparison does not systematically favor either feasible action and provides an action-independent measure of structural-fidelity improvement. This criterion characterizes the local conditions under which selective evolution is expected to be beneficial, rather than implying global optimality of the learned policy.

4.3.4. Joint adaptive training

The total training objective of RE-DGCL is a weighted combination of the hierarchical consistency loss and the decisional policy loss:

$$\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{con}} + \alpha\mathcal{L}_{\text{RL}} \tag{17}$$

where α is a trade-off coefficient. This joint optimization creates a self-correcting learning loop: the contrastive objective improves the representation space, while the jointly optimized SGE branch provides a progressively refined structural reference for counterfactual policy evaluation.

5. Experiments

This section evaluates the performance of RE-DGCL on eight real-world datasets for the dynamic link prediction task. An empirical validation of the "Hard Evolutionary Boundary" problem is first presented to justify the model's motivation. RE-DGCL is then compared against 12 representative baselines to evaluate its overall effectiveness, followed by a systematic ablation study and hyperparameter sensitivity analysis to assess the contribution of each core module.

Table 2
Statistics of the datasets.

Datasets	Nodes	Edges	Timespan(days)	Sparsity Level
UCI	1899	59,835	193.71	High
Enron	184	125,235	1,316.44	Moderate
Reddit	10,984	672,447	46.5	Extreme
MOOC	7144	411,749	29.77	Extreme
Fb-forum	899	33,720	164.49	High
BITalpha	3783	24,186	1,901.00	Extreme
UNtrade	255	507,497	11,323.00	Moderate
LastFM	1980	1,293,103	1,586.89	High

5.1. Experimental settings

5.1.1. Datasets

To comprehensively evaluate the generalization capability and robustness of RE-DGCL across diverse dynamic scenarios, experiments are conducted on eight real-world datasets exhibiting significant differences in network scale, interaction density, temporal span, and generation mechanism. Table 2 presents the key statistical characteristics of all datasets, categorized by their structural and temporal sparsity.

- **UCI (Panzarasa et al., 2009)**: A directed social network derived from the online community of students at the University of California, Irvine. Nodes represent community users and edges denote timestamped message-sending behaviors, reflecting typical campus social interaction patterns with moderate interaction density.
- **Enron (Shetty & Adibi, 2004)**: Originates from Enron's publicly released email corpus. Nodes represent company employees and edges denote timestamped email communications. With an extensive time span of approximately 3.5 years, this dataset is frequently used to analyze the long-term evolution of internal organizational communication structures, exhibiting pronounced historical dependency.
- **Reddit (Kumar et al., 2019b)**: A bipartite interaction graph sourced from Reddit. Nodes consist of users and subreddits, and edges represent posting or commenting activities. This dataset exhibits extreme snapshot-level sparsity, where active edges at each timestamp account for less than 0.1% of all possible node pairs, posing significant challenges to traditional forced-update dynamic GNNs.
- **MOOC (Kumar et al., 2019b)**: Contains user behavior data from Massive Open Online Course platforms. Nodes include students and course objectives such as videos and assignments, while edges represent specific learning interactions. This network captures users' temporal behavioral patterns and attrition trends within educational contexts.
- **Fb-forum (Opsahl & Panzarasa, 2009)**: An activity network based on Facebook forum users. Nodes represent users and edges correspond to interactions or replies under similar topics. Though relatively small in scale, this dataset exhibits a dense social interaction structure with high interaction frequency.
- **BITalpha (Kumar et al., 2016)**: A Bitcoin transaction trust network in which nodes represent traders and edges denote timestamped trust evaluations or transaction behaviors. As a financial network spanning nearly 1901 days, it contains critical temporal signals regarding trust establishment and evolution, with extreme structural sparsity.
- **UNtrade (MacDonald et al., 2015)**: Sourced from the United Nations Commodity Trade Statistics Database, recording annual trade flows between nations. Nodes represent countries and edges represent import/export relationships. Spanning over 30 years, this dataset tests a model's ability to capture long-term macroeconomic structural evolution.
- **LastFM (Kumar et al., 2019b)**: Sourced from the Last.fm online music platform. Nodes represent users and artists, and edges indicate listening events, reflecting the dynamic evolution of music preferences over extended periods.

5.1.2. Baselines

RE-DGCL is compared against 12 state-of-the-art baseline methods spanning four categories, ensuring comprehensive coverage of the dynamic graph representation learning landscape.

- **Static graph methods**: GraphSAGE (Hamilton et al., 2017) proposes an inductive framework that learns node representations by sampling and aggregating features from local neighborhoods. DGI (Veličković et al., 2018) maximizes mutual information between local patch representations and a global graph summary, serving as a foundational self-supervised baseline.
- **Supervised continuous-time dynamic graph methods**: JODIE (Kumar et al., 2019a) employs two mutually coupled RNNs to update embedding trajectories and introduces a projection operator that predicts future embedding states. DyRep (Trivedi et al., 2019) models dynamic graph evolution through a deep temporal point process model that distinguishes topological evolution from node interaction. TGN (Rossi et al., 2020) is the current benchmark for continuous-time dynamic graph learning, combining an explicit memory module with a graph attention mechanism for excellent inductive performance.
- **Discrete-time and attention-based methods**: DySAT (Sankar et al., 2020) employs structural and temporal self-attention to capture both local neighborhood features and evolutionary patterns across snapshots. GraphMixer (Cong et al., 2023) proposes a simplified MLP-Mixer-based architecture that challenges the necessity of complex recurrent structures. DyGFormer (Yu et al., 2023) is a Transformer-based model that learns from first-order historical interaction sequences through neighbor co-occurrence encoding and a patching technique for long-term dependency modeling. MTdyg (Xu et al., 2024a) introduces a multi-scale patching strategy with Fourier-based adaptive patch size selection combined with temporal attention.
- **Self-supervised and contrastive learning methods**: DGMSCL (Qian et al., 2025) proposes a hybrid supervised contrastive learning framework using hierarchical attention graph neural networks. CPDG (Bei et al., 2023) leverages temporal neighborhood signals as self-supervised objectives, enabling transferable representations across downstream tasks. DMetaGCRN (Guo et al., 2025) dynamically constructs heterogeneous-aware graph structures to simultaneously capture spatial correlations and temporal dependencies.

All baseline models adopt parameter settings from their original papers or official code repositories. All methods are evaluated under the unified protocol described in Section 5.1.4 to ensure experimental fairness.

5.1.3. Evaluation metrics

Model performance is evaluated using two widely adopted standard metrics: Area Under the ROC Curve (AUC) (Fawcett, 2006) and Average Precision (AP) (Kumaran & Allan, 2008). Both metrics are threshold-independent, reflecting the model's discriminative ability from complementary perspectives.

- **AUC** quantifies the probability that the model ranks a randomly selected positive sample (an existing link) above a randomly selected negative sample (a non-existent link). A higher AUC indicates stronger ranking quality and overall discriminative power.
- **AP** is the area under the Precision-Recall (PR) curve, summarizing the weighted average precision across different recall levels. Since real-world dynamic graphs are typically extremely sparse—where negative samples vastly outnumber positive samples—AP is considered a more informative and rigorous metric in such class-imbalanced scenarios.

5.1.4. Evaluation protocol

We adopt a chronological one-step-ahead evaluation protocol for dynamic link prediction. For each dataset, the interaction sequence is first

partitioned into consecutive graph snapshots according to the dataset-specific temporal window. The first T_{train} snapshots are used exclusively for representation learning, while the immediately subsequent snapshot is employed as the target snapshot for downstream link prediction. Specifically, T_{train} is set to 6, 9, 8, 8, 5, 9, 8, and 8 for UCI, Enron, Reddit, MOOC, Fb-forum, BITalpha, UNtrade, and LastFM, respectively. Consequently, no structural information from the target snapshot is used during representation learning.

For downstream evaluation, observed edges in the target snapshot are regarded as positive samples, while an equal number of unconnected node pairs are randomly sampled as negative samples. The resulting positive and negative edge pairs are randomly divided into training, validation, and test subsets with proportions of 20%, 20%, and 60%, respectively. For each node pair, the edge representation is constructed using the Hadamard product of the corresponding node embeddings, and a logistic regression classifier is trained on the training subset. Validation AUC is used for model selection, while AUC and AP on the held-out test subset are reported as the final evaluation metrics.

All experiments are conducted under a transductive setting, where the node set is known during representation learning but the links in the target prediction snapshot remain unavailable to the representation encoder. The same evaluation protocol and negative-sampling procedure are applied to all compared methods to ensure fairness. Each experiment is independently repeated five times using different random seeds. The main comparison tables report the average performance over the five runs, while the standard deviations of RE-DGCL are additionally provided in the following result analysis to evaluate its experimental stability.

5.1.5. Parameter settings

All experiments were conducted on a high-performance Linux server equipped with an Intel(R) Xeon(R) Gold 6530 CPU (16 cores), an NVIDIA GeForce RTX 4090 GPU (24 GB VRAM), and 127 GB of memory. The model was implemented using Python 3.10 with the PyTorch 1.12.1 framework and CUDA 11.3. The Adam optimizer is employed for parameter refinement, and early stopping with a patience of 100 epochs is applied to prevent overfitting. For efficiency evaluation, the per-epoch training time is averaged over 10 epochs after excluding the initial warm-up epochs, and peak GPU memory consumption is recorded during model training under the same hardware environment.

The learning rate (lr) was searched within $\{0.0005, 0.001, 0.002, 0.005, 0.008\}$ and the output embedding dimension (out_dim) was selected from $\{32, 64, 128\}$. Dropout mechanisms are incorporated into both the feature layers and attention modules to mitigate overfitting.

Table 3
Optimal parameter configurations for each dataset.

Parameters	UCI	Enron	Reddit	MOOC	Fb-forum	BITalpha	UNtrade	LastFM
Dropout-feat	0.3	0.5	0.2	0.2	0.2	0.4	0.5	0.2
Dropout-attn	0.2	0.3	0.6	0.5	0.5	0.6	0.2	0.5
out_dim	64	32	128	128	64	64	32	128
lr	0.005	0.001	0.002	0.005	0.008	0.005	0.0005	0.005
τ	0.2	0.5	0.2	0.2	0.5	0.1	0.5	0.2
K	5	2	10	8	3	12	2	6
δ	0.2	0.5	0.4	0.1	0.3	0.3	0.6	0.2
λ_0	0.05	0.1	0.15	0.05	0.05	0.2	0.1	0.05
α	0.15	0.1	0.15	0.15	0.15	0.2	0.05	0.15
η	0.005	0.005	0.01	0.01	0.005	0.005	0.001	0.005

The feature dropout rate ($Dropout\text{-}feat$) was tuned within $[0.2, 0.5]$, and the attention dropout rate ($Dropout\text{-}attn$) within $[0.2, 0.6]$. The temperature parameter τ was fine-tuned within $[0.1, 0.5]$. The diffusion step count K was selected from $[2, 12]$ based on dataset sparsity and evolutionary scale. The loss balance coefficient δ was searched within $[0.1, 0.7]$. The initial difficulty ratio λ_0 and the RL loss weight α were optimized within $[0.05, 0.2]$ and $[0.01, 0.2]$, respectively. The final optimal configurations, determined via validation set evaluation, are detailed in Table 3.

5.1.6. Motivation validation: The hard evolutionary boundary problem

Before presenting the main experimental results, the core motivation of RE-DGCL is empirically validated by demonstrating that the Hard Evolutionary Boundary problem is a genuine and pervasive challenge in real-world dynamic graphs.

Temporal Characteristics of Dynamic Graphs. Two representative temporal properties are quantified following prior work (Gao et al., 2024): historical dependency Ψ_H, which measures the average proportion of recurring edges at each timestamp, and temporal proximity preference Ψ_T, which captures the degree to which nodes prefer to reconnect with recently interacted neighbors. As illustrated in Fig. 4, all eight datasets exhibit both Ψ_H and Ψ_T values substantially above 0.5, indicating that these temporal biases are consistently observed across diverse dynamic network domains. Specifically, Ψ_H ranges from 0.53 (UCI) to 0.89 (UNtrade), reflecting varying degrees of historical interaction recurrence. UNtrade achieves the highest Ψ_H of 0.89, consistent with the highly stable and slowly evolving nature of international trade partnerships spanning decades. MOOC and Enron also exhibit high Ψ_H values of 0.88 and 0.86 respectively, indicating that recurring interaction patterns are dominant even in educational and organizational communication networks. Meanwhile, Ψ_T values range from 0.73 (Fb-forum)

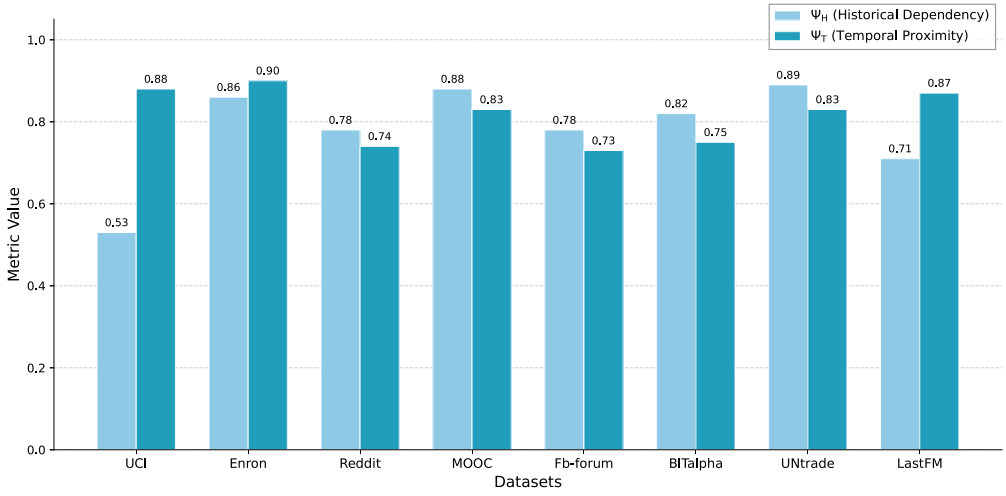


Fig. 4. Temporal characteristics of dynamic graphs.

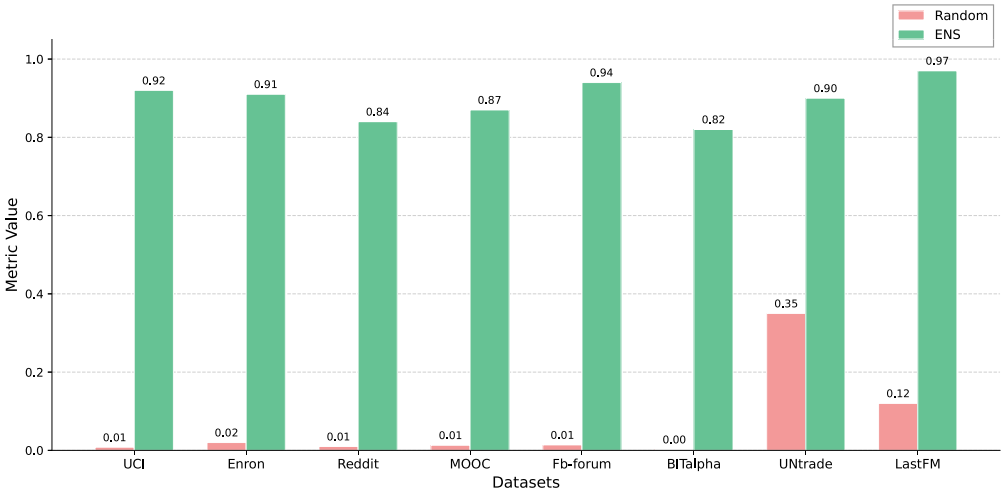


Fig. 5. Negative sample quality comparison between random sampling and ENS.

to 0.90 (Enron), with Enron exhibiting the strongest temporal proximity preference—reflecting the concentrated and time-sensitive nature of workplace email communications. Notably, even datasets dominated by high-frequency short-burst interactions such as Reddit ($\Psi_H = 0.78$, $\Psi_T = 0.74$) and Fb-forum ($\Psi_H = 0.78$, $\Psi_T = 0.73$) maintain both metrics well above 0.5. These findings collectively indicate that nodes which previously interacted but are currently disconnected are semantically more similar to positive samples than nodes that have never interacted, providing an empirical basis for the Hard Evolutionary Boundary problem.

Failure of Random Negative Sampling. Despite the prevalence of these temporal biases, many existing dynamic GCL methods rely on random uniform sampling to construct the negative semantic space. As demonstrated in Fig. 5, random negative sampling retrieves historical edges at rates of merely 0.00–0.35 across all eight datasets. In six out of eight datasets, this rate falls at or below 0.02, meaning that over 98% of randomly selected negatives are nodes that have never interacted with the anchor node—trivially distinguishable "easy negatives" that provide no meaningful contrastive gradient for learning evolutionary boundaries. The situation is most severe on BITalpha, where the random retrieval rate reaches 0.00 due to extreme structural sparsity, meaning random sampling effectively never encounters any historically interacted node. In contrast, the proposed ENS module consistently identifies between 82% and 97% of historically relevant nodes as hard negatives across all datasets. The contrast is particularly striking on BITalpha (0.00 vs. 0.82) and LastFM (0.12 vs. 0.97), where ENS achieves substantially higher retrieval rates than random sampling by leveraging the history interaction buffer. On UNtrade, where random sampling achieves a relatively higher retrieval rate of 0.35 due to its denser cumulative edge set, ENS still substantially improves upon it with a rate of 0.90. Even on Reddit and MOOC, where random sampling achieves only 0.01, ENS reliably retrieves 0.84 and 0.87 of historical edges respectively.

These empirical results primarily validate the motivation for history-aware negative selection. Standard random sampling rarely retrieves temporally relevant historical neighbors, particularly in sparse dynamic graphs, whereas ENS effectively identifies substantially more informative candidate constraints. However, the retrieval analysis alone does not isolate the contribution of curriculum pacing. We therefore provide an additional fine-grained ablation in Section 5.3.3 to separately evaluate history-aware negative selection and progressive difficulty scheduling. Together, these observations provide rigorous data-driven justification for the ENS design and explain why RE-DGCL achieves particularly pronounced improvements on datasets with high historical dependency and extreme sparsity, as demonstrated in Section 5.2.

5.2. Overall performances

To validate the effectiveness of RE-DGCL in dynamic link prediction, Tables 4 and 5 report the AUC and AP performance of RE-DGCL alongside 12 state-of-the-art algorithms across eight datasets. The reported RE-DGCL results are averaged over five independent runs with different random seeds. The best performance is indicated in bold and the second best is underlined. Overall, RE-DGCL achieves the highest AUC on all eight datasets and the highest AP on seven datasets, while ranking second in AP on *UCI*. Its average AUC and AP reach 0.9370 and 0.9178, respectively. Compared with the strongest competing result on each dataset, RE-DGCL yields average improvements of approximately 0.0176 in AUC and 0.0143 in AP. The consistent gains over strong self-supervised baselines such as CPDG and DGMSCL, as well as Transformer-based models such as DyGFormer, demonstrate the generalization capability of the proposed proactive decisional evolution paradigm across heterogeneous dynamic scenarios.

At the dataset level, RE-DGCL exhibits particularly clear advantages on *Enron*, *BITalpha*, and *UNtrade*. On *Enron*, RE-DGCL achieves AUC/AP scores of 0.9381/0.9214, outperforming the second-best CPDG by 0.0183/0.0180. On *BITalpha*, it obtains 0.9442/0.9181, exceeding MTdyg by 0.0270/0.0238. Similarly, RE-DGCL reaches 0.8996/0.8835 on *UNtrade*, compared with CPDG's 0.8812/0.8634. These results indicate that methods relying mainly on passive sequential modeling may struggle to preserve reliable structural information under irregular dynamic evolution. In contrast, the EDA mechanism selectively controls state updates and reduces the influence of uninformative evolutionary signals.

On datasets containing frequent interaction streams, RE-DGCL also maintains competitive advantages. It achieves 0.9508/0.9347 on *Reddit*, surpassing the second-best CPDG by 0.0130/0.0102. On *LastFM*, RE-DGCL obtains the highest AUC of 0.9402, improving over MTdyg's 0.9311, while their AP results are nearly identical at 0.9154 and 0.9152. This narrower gap indicates that strong temporal sequence models can already capture substantial information when interactions are highly continuous. Nevertheless, RE-DGCL retains an overall advantage by combining selective state evolution with ENS-based contrastive optimization.

On *UCI*, RE-DGCL achieves the highest AUC of 0.9594, exceeding DGMSCL's 0.9502. Its AP of 0.9461 ranks second to DyGFormer's 0.9584. This result shows that RE-DGCL remains highly competitive on relatively stable interaction patterns, although DyGFormer exhibits an advantage in AP on this dataset. On *MOOC*, RE-DGCL obtains 0.9215/0.8970, improving over the strongest competing results by 0.0260 in AUC and 0.0358 in AP.

Table 4
Performance comparison (Part 1).

datasets	UCI		Enron		Reddit		MOOC	
metrics	auc	ap	auc	ap	auc	ap	auc	ap
DGI	0.8186	0.7866	0.7277	0.6918	0.7921	0.7583	0.5565	0.5371
GraphSAGE	0.7557	0.7213	0.6829	0.6517	0.7213	0.7031	0.6256	0.6012
JODIE	0.8729	0.8404	0.7931	0.7405	0.8137	0.8066	0.6533	0.6200
DyRep	0.7324	0.6816	0.6605	0.6301	0.8291	0.8013	0.8014	0.7739
TGN	0.8808	0.8484	0.7663	0.7234	0.7706	0.7513	0.8670	0.8518
DySAT	0.8956	0.8845	0.9021	0.8933	0.8089	0.7713	0.8488	0.8237
GraphMixer	0.8537	0.8016	0.8456	0.8043	0.9011	0.8862	0.8074	0.7832
DyGFormer	0.9459	**0.9584**	0.9028	0.8994	0.9327	0.9123	0.8077	0.8586
MTdyg	0.9253	0.9029	0.9069	0.8821	0.9320	0.9237	0.8426	0.8266
DGMSCL	0.9502	0.8991	0.8644	0.8010	0.8655	0.8510	0.8629	0.8507
DMetaGCRN	0.9058	0.8506	0.7935	0.8567	0.9015	0.8851	0.8955	0.8402
CPDG	0.9312	0.9187	0.9198	0.9034	0.9378	0.9245	0.8734	0.8612
RE-DGCL	**0.9594**	0.9461	**0.9381**	**0.9214**	**0.9508**	**0.9347**	**0.9215**	**0.8970**
w/o EDA	0.9378	0.9234	0.9056	0.8867	0.9312	0.9145	0.8978	0.8712
w/o ENS	0.9424	0.9281	0.9101	0.8914	0.9358	0.9197	0.9055	0.8790
w/o TE	0.9462	0.9317	0.9187	0.8991	0.9344	0.9178	0.9038	0.8783
w/o GD	0.9478	0.9341	0.9216	0.9026	0.9363	0.9204	0.9047	0.8796
w/o EFR	0.9410	0.9270	0.9152	0.8964	0.9326	0.9163	0.9023	0.8774

Table 5
Performance comparison (Part 2).

datasets	Fb-forum		BITalpha		UNtrade		LastFM	
metrics	auc	ap	auc	ap	auc	ap	auc	ap
DGI	0.8118	0.7798	0.8118	0.7819	0.7236	0.7011	0.7512	0.7360
GraphSAGE	0.7465	0.7053	0.7065	0.6735	0.5927	0.5613	0.5322	0.5170
JODIE	0.8012	0.7823	0.7933	0.7627	0.6918	0.6384	0.6900	0.6938
DyRep	0.7592	0.7343	0.8274	0.7998	0.8295	0.7914	0.6841	0.6726
TGN	0.8678	0.8459	0.8375	0.8067	0.6637	0.6223	0.5612	0.5712
DySAT	0.8718	0.8535	0.8617	0.8213	0.8333	0.8182	0.8009	0.7817
GraphMixer	0.8721	0.8554	0.8391	0.8124	0.6658	0.6013	0.7460	0.7651
DyGFormer	0.9066	0.8942	0.8617	0.8434	0.8775	0.8547	0.8923	0.8850
MTdyg	0.9218	0.9080	0.9172	0.8943	0.6773	0.6821	0.9311	0.9152
DGMSCL	0.8780	0.8623	0.8787	0.8321	0.8619	0.8236	0.8360	0.8482
DMetaGCRN	0.8216	0.8113	0.7839	0.7920	0.7279	0.7368	0.7646	0.7663
CPDG	0.9134	0.8978	0.9023	0.8867	0.8812	0.8634	0.9089	0.8934
RE-DGCL	**0.9418**	**0.9265**	**0.9442**	**0.9181**	**0.8996**	**0.8835**	**0.9402**	**0.9154**
w/o EDA	0.9167	0.8989	0.8934	0.8712	0.8623	0.8434	0.9212	0.8978
w/o ENS	0.9248	0.9082	0.9187	0.8954	0.8795	0.8607	0.9172	0.8932
w/o EFR	0.9286	0.9109	0.9204	0.8910	0.8732	0.8548	0.9259	0.9010
w/o TE	0.9302	0.9145	0.9279	0.9031	0.8845	0.8673	0.9298	0.9060
w/o GD	0.9227	0.9054	0.9058	0.8827	0.8710	0.8519	0.9286	0.9039

In summary, RE-DGCL achieves consistent AUC improvements across all evaluated datasets and leading AP performance on most datasets. Its advantages are especially evident on datasets with irregular or complex evolution, while it remains competitive under continuous and relatively stable interaction patterns. These results demonstrate the consistent effectiveness and applicability of the proposed framework across heterogeneous dynamic graphs.

In addition, the statistical stability of RE-DGCL is evaluated over five independent runs with different random seeds. The corresponding AUC/AP standard deviations are 0.0021/0.0028 on UCI, 0.0034/0.0040 on Enron, 0.0027/0.0031 on Reddit, 0.0042/0.0048 on MOOC, 0.0030/0.0036 on Fb-forum, 0.0041/0.0047 on BITalpha, 0.0033/0.0039 on UNtrade, and 0.0029/0.0030 on LastFM. The consistently small variations across all eight datasets indicate that RE-DGCL maintains stable performance under different random initializations.

5.3. Ablation study

5.3.1. Component ablation

To rigorously evaluate the individual contributions of RE-DGCL's core components, a series of ablation studies are conducted by progressively removing or substituting key modules. Five representative variants are constructed as follows:

- **RE-DGCL w/o EDA**: The RL-driven Evolutionary Decision Action mechanism is deactivated and replaced with a traditional forced-update strategy that unconditionally updates node states at each evolutionary step. This variant evaluates the overall benefit of selective state evolution over unconditional updating.
- **RE-DGCL w/o ENS**: The Evolutionary-aware Negative Sampling module is removed and replaced with standard random uniform sampling. This ablation evaluates whether the proposed hard negative constraints successfully sharpen the contrastive boundary and enhance discriminative power in dynamic environments.
- **RE-DGCL w/o EFR**: The counterfactual Evolutionary Fidelity Reward is replaced with an action-independent constant reward, preventing the policy from distinguishing the relative utility of evolutionary updating and state maintenance. This variant evaluates the necessity of counterfactual reward feedback for learning an effective update policy.
- **RE-DGCL w/o GD**: The graph diffusion module is removed from the global encoder, replacing it with direct multi-head attention on raw node features without spatial smoothing. This experiment investigates the role of the "diffuse-first, aggregate-later" strategy in capturing stable macro-community structures and filtering micro-level topology perturbations.

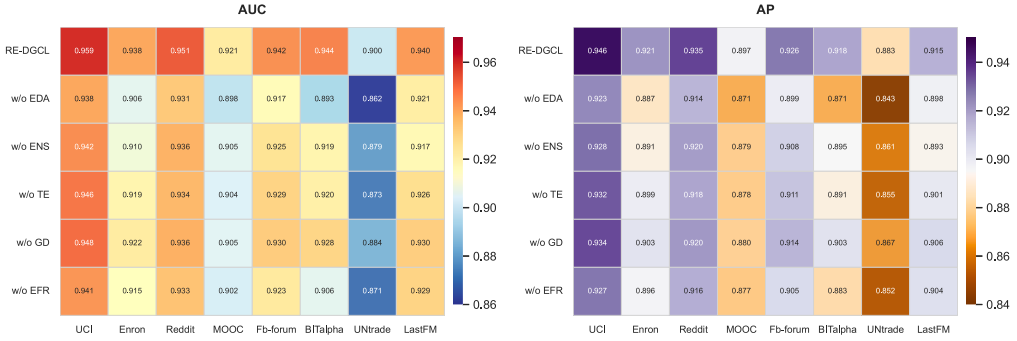


Fig. 6. Ablation study results (AUC and AP) across eight datasets.

- **RE-DGCL w/o TE**: The harmonic temporal encoding module is removed from the local encoder, employing standard spatial aggregation without temporal frequency decomposition. This variant evaluates the importance of fine-grained periodicity and time-interval encoding for capturing high-frequency interaction dynamics.

The performance of these variants across eight datasets is visualized as heatmaps in Fig. 6, with detailed results summarized in the lower section of Tables 4 and 5.

Several critical observations emerge from the ablation results. First, removing the EDA mechanism (**w/o EDA**) causes the largest average degradation across the eight datasets, especially on *BITalpha* and *UNtrade*. On *BITalpha*, the AUC/AP decreases from 0.9442/0.9181 to 0.8934/0.8712, while on *UNtrade*, it drops from 0.8996/0.8835 to 0.8623/0.8434. These results demonstrate the importance of selective state evolution over unconditional updating, since indiscriminate state transitions may propagate unreliable interaction signals.

Second, the **w/o ENS** variant produces consistent performance losses across all datasets, with particularly clear degradation on *Enron*, *BITalpha*, *UNtrade*, and *LastFM*. This indicates that random sampling provides less informative contrastive constraints, whereas evolutionary hard negatives help establish more discriminative relationship boundaries.

Third, the **w/o EFR** variant also exhibits consistent degradation, with relatively pronounced losses on *Enron*, *BITalpha*, and *UNtrade*. This confirms that an action-independent constant reward cannot provide discriminative feedback regarding the relative utility of evolutionary updating and state maintenance, thereby weakening the learned gating policy.

Finally, removing either Graph Diffusion (GD) or Temporal Encoding (TE) leads to moderate but consistent performance decay. The **w/o GD** variant produces comparatively large AUC reductions on *BITalpha* and *UNtrade*, whereas **w/o TE** shows clearer losses on *Enron*, *MOOC*, and *Reddit*. These findings validate the complementary roles of macro-structural diffusion and fine-grained temporal modeling.

In summary, the component ablation demonstrates that each major module contributes positively to the overall performance of RE-DGCL, while their joint optimization provides complementary benefits for dynamic representation learning. In particular, the integration of selective decisional evolution, counterfactual reward feedback, and evolutionary negative sampling improves both the stability and discriminative capability of the learned representations.

5.3.2. Comparison with alternative update policies

Although the component ablation demonstrates the benefit of selective state evolution, comparison with forced updating alone cannot determine whether the improvement originates specifically from reinforcement learning or more generally from selective memory updating. We therefore compare EDA with two simpler update strategies while keeping all other components unchanged.

Random Gate. State updates are sampled using a fixed Bernoulli probability matched to the average update ratio of RE-DGCL, controlling for reduced update frequency without learned decision making.

Deterministic Gate. The discrete RL action is replaced with a continuously differentiable gate $p_{i,t} \in [0, 1]$, producing the ALE state as

$$Z_{i,t}^{\mathrm{ALE}} = p_{i,t} Z_{i,t}^{U} + (1 - p_{i,t}) Z_{i,t}^{M}, \tag{18}$$

The gating network is optimized directly through the representation-learning objective without policy-gradient optimization, thereby retaining learned selective updating while removing the RL formulation. For a fair comparison, the Deterministic Gate adopts the same gating-network architecture as EDA, while all remaining model components and hyperparameters are kept unchanged across all alternative update strategies.

The comparison is conducted on Fb-forum, BITalpha, and UNtrade, which exhibit diverse interaction densities and temporal evolution patterns.

As shown in Table 6, selective-update strategies generally outperform unconditional forced updating, confirming the benefit of suppressing unnecessary state transitions. Random gating provides relatively limited gains, indicating that reducing the update frequency alone is insufficient, while the deterministic gate further benefits from task-adaptive state selection. The comparison with RL-driven EDA further distinguishes continuous representation mixing from discrete update-or-maintain decisions, indicating that both selective evolution and the specific decision mechanism contribute to the overall performance.

5.3.3. Fine-grained analysis of ENS

To disentangle history-aware negative selection from curriculum pacing, we compare random negative sampling, HNC sampling without curriculum, and the full curriculum-based ENS. In the HNC w/o Curriculum variant, historical hard negatives are retained but introduced with a fixed sampling ratio throughout training.

As shown in Table 7, history-aware HNC sampling consistently outperforms random negative sampling, confirming that temporally relevant historical neighbors provide more informative contrastive signals than uniformly sampled easy negatives. Introducing curriculum pacing yields further improvements, indicating that history-aware negative identification and progressive difficulty scheduling play complementary roles: the former determines which negatives are informative, while the latter controls when and to what extent these difficult constraints participate in training.

Nevertheless, HNCs remain snapshot-conditional constraints, and a subset of historically related nodes may reconnect in future snapshots. Curriculum pacing helps reduce the early influence of such ambiguous pairs, while completely resolving future-reconnection-induced false negatives remains an important direction for future work.

5.4. Robustness under random edge perturbations

To directly evaluate robustness against stochastic interaction noise, we inject $\lfloor \rho | E_t | \rfloor$ random edges into each training snapshot, where

Table 6
Comparison of different state-update strategies on representative datasets.

Update Strategy	Fb-forum		BITalpha		UNtrade	
	AUC	AP	AUC	AP	AUC	AP
Forced Update	0.9167 ± 0.0045	0.8989 ± 0.0051	0.8934 ± 0.0058	0.8712 ± 0.0064	0.8623 ± 0.0049	0.8434 ± 0.0055
Random Gate	0.9234 ± 0.0041	0.9060 ± 0.0048	0.9025 ± 0.0052	0.8801 ± 0.0058	0.8721 ± 0.0045	0.8528 ± 0.0050
Deterministic Gate	0.9348 ± 0.0035	0.9189 ± 0.0040	0.9326 ± 0.0046	0.9067 ± 0.0051	0.8928 ± 0.0039	0.8764 ± 0.0044
RL-driven EDA	0.9418 ± 0.0030	0.9265 ± 0.0036	0.9442 ± 0.0041	0.9181 ± 0.0047	0.8996 ± 0.0033	0.8835 ± 0.0039

Table 7
Fine-grained comparison of negative-sampling strategies on representative datasets.

Negative Sampling	Fb-forum		BITalpha		UNtrade	
	AUC	AP	AUC	AP	AUC	AP
Random Negative	0.9248 ± 0.0040	0.9082 ± 0.0047	0.9187 ± 0.0050	0.8954 ± 0.0056	0.8795 ± 0.0044	0.8607 ± 0.0050
HNC w/o Curriculum	0.9356 ± 0.0035	0.9199 ± 0.0041	0.9368 ± 0.0045	0.9109 ± 0.0050	0.8920 ± 0.0039	0.8751 ± 0.0044
Curriculum ENS	0.9418 ± 0.0030	0.9265 ± 0.0036	0.9442 ± 0.0041	0.9181 ± 0.0047	0.8996 ± 0.0033	0.8835 ± 0.0039

Table 8
Robustness comparison under random edge perturbations on three representative datasets. The reported results are averaged over five independent runs.

Method	Noise Ratio	Fb-forum		BITalpha		UNtrade	
		AUC	AP	AUC	AP	AUC	AP
TGN	0%	0.8678	0.8459	0.8375	0.8067	0.6637	0.6223
TGN	10%	0.8469	0.8226	0.8104	0.7778	0.6458	0.6021
TGN	20%	0.8206	0.7935	0.7776	0.7421	0.6234	0.5770
DyGFormer	0%	0.9066	0.8942	0.8617	0.8434	0.8775	0.8547
DyGFormer	10%	0.8924	0.8785	0.8428	0.8228	0.8622	0.8378
DyGFormer	20%	0.8735	0.8573	0.8184	0.7963	0.8444	0.8181
CPDG	0%	0.9134	0.8978	0.9023	0.8867	0.8812	0.8634
CPDG	10%	0.9021	0.8854	0.8871	0.8702	0.8690	0.8499
CPDG	20%	0.8872	0.8688	0.8676	0.8483	0.8543	0.8335
RE-DGCL	0%	**0.9418**	**0.9265**	**0.9442**	**0.9181**	**0.8996**	**0.8835**
RE-DGCL	10%	**0.9326**	**0.9163**	**0.9330**	**0.9058**	**0.8921**	**0.8752**
RE-DGCL	20%	**0.9205**	**0.9029**	**0.9194**	**0.8906**	**0.8823**	**0.8644**

$\rho \in \{0\%, 10\%, 20\%\}$. Self-loops, duplicate or existing edges, and positive edges in the target snapshot are excluded. The target snapshot remains unchanged to prevent information leakage. RE-DGCL, TGN, DyGFormer, and CPDG are evaluated on the same perturbed snapshots of Fb-forum, BITalpha, and UNtrade over five independent runs.

As shown in Table 8, all methods degrade as the perturbation ratio increases, while RE-DGCL consistently achieves the best AUC and AP. From 0% to 20% noise, its AUC decreases by 2.13, 2.48, and 1.73 percentage points on Fb-forum, BITalpha, and UNtrade, respectively, compared with 4.72, 5.99, and 4.03 points for TGN. DyGFormer and CPDG exhibit intermediate degradation. These results indicate that stable structural anchoring and selective state evolution help limit the propagation of randomly injected interactions, although extremely sparse graphs such as BITalpha remain more sensitive to perturbations.

5.5. Hyperparameter sensitivity analysis

To evaluate the stability and adaptability of RE-DGCL, we analyze the sensitivity of four key hyperparameters: the graph diffusion step count K, the loss-balancing coefficient δ, the ENS initial difficulty ratio λ_0, and the RL loss weight α. Experiments are conducted on five representative datasets, namely UCI, Enron, MOOC, BITalpha, and UNtrade, with the results reported in Figs. 7–9.

5.5.1. Impact of graph diffusion steps K

Fig. 7 presents the sensitivity of RE-DGCL to the graph diffusion depth $K \in \{1, \ldots, 18\}$. Overall, AUC and AP generally increase at shallow diffusion depths and gradually decline when K becomes excessively large. A small K restricts the global encoder to local neighborhoods and may provide insufficient higher-order structural information, whereas deep diffusion propagates information from increasingly distant nodes, potentially introducing irrelevant signals and over-smoothing.

The optimal diffusion depth varies substantially across datasets. *Enron* and *UNtrade* reach their best AUC/AP values at $K = 2$, indicating that their useful structural information can be captured through relatively shallow propagation. In contrast, *UCI* and *MOOC* achieve their optima at $K = 5$ and $K = 8$, respectively, suggesting that moderate multi-hop diffusion is beneficial for extracting broader structural context. *BITalpha* requires the largest depth, reaching its highest AUC/AP of 0.9442/0.9181 at $K = 12$, which indicates a stronger dependence on higher-order structural relationships. Beyond these optimal points, both metrics generally decline due to accumulated noise and excessive smoothing.

Therefore, the optimal values are $K = 5$, 2, 8, 12, and 2 for *UCI*, *Enron*, *MOOC*, *BITalpha*, and *UNtrade*, respectively. The consistent peak positions of AUC and AP support the reliability of these settings and confirm that diffusion depth should be selected according to dataset-specific structural characteristics.

5.5.2. Impact of δ on loss balance

Fig. 8 reports the influence of the loss-balancing coefficient δ, which is varied from 0.1 to 0.9. AUC and AP exhibit similar variation patterns on each dataset and reach their maxima at the same value of δ, although the optimal setting differs across datasets. This demonstrates that the relative contribution of the associated optimization objectives should be adjusted according to different dynamic environments.

UCI obtains its best performance at $\delta = 0.2$, while *BITalpha* reaches its optimum at $\delta = 0.3$. Both datasets favor relatively small or moderate coefficients, after which performance decreases as the auxiliary objective receives excessive emphasis. *Enron* achieves its highest AUC/AP of 0.9381/0.9214 at $\delta = 0.5$, indicating that a more balanced contribution of the loss components is beneficial. *MOOC* performs best at the smallest tested value, $\delta = 0.1$, and generally deteriorates as δ increases. In contrast, *UNtrade* favors a larger coefficient and reaches its optimum at $\delta = 0.6$, with competitive results also observed around $\delta = 0.5$–0.7.

In summary, the optimal values of δ are 0.2, 0.5, 0.1, 0.3, and 0.6 for *UCI*, *Enron*, *MOOC*, *BITalpha*, and *UNtrade*, respectively. These results indicate that an excessively large coefficient may interfere with the dominant learning objective, whereas an overly small value may provide insufficient auxiliary supervision. The settings reported in Table 3 therefore follow the corresponding peak positions in Fig. 8.

5.5.3. Impact of evolvable components (λ_0 and α)

We further investigate two hyperparameters associated with the evolvable mechanisms of RE-DGCL: the ENS initial difficulty ratio λ_0

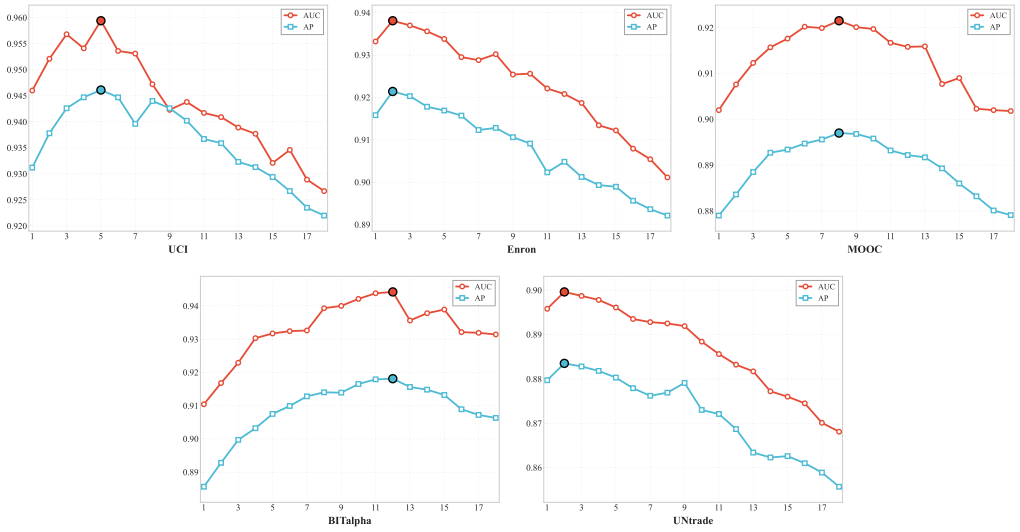


Fig. 7. Impact of graph diffusion steps K on model performance.

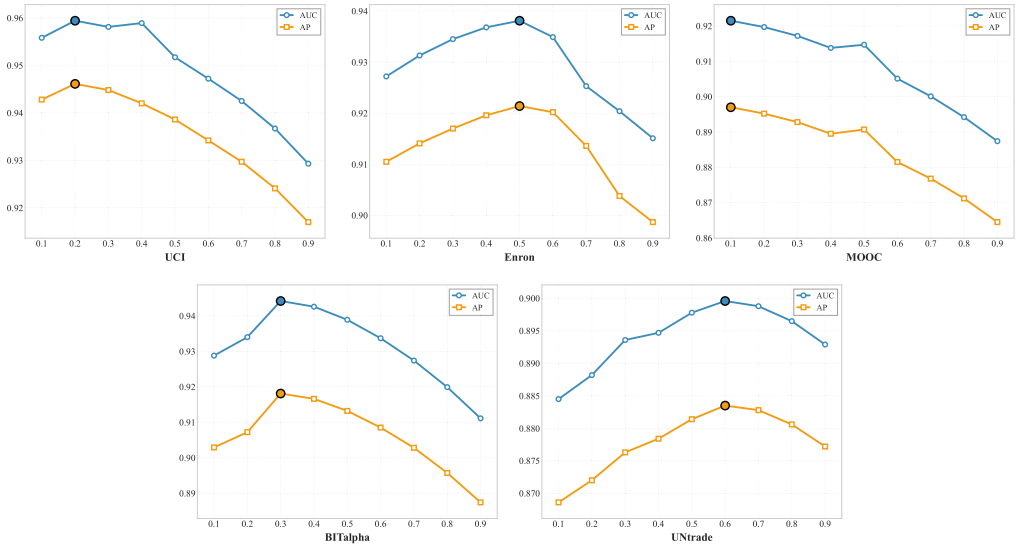


Fig. 8. Impact of the loss-balancing coefficient δ on model performance.

and the RL loss weight α. The corresponding results are presented in Fig. 9.

Impact of ENS Initial Difficulty Ratio λ_0. Fig. 9 (left) shows that most datasets exhibit an initial improvement followed by a decline as λ_0 increases. When λ_0 is too small, the curriculum begins with insufficient informative hard negatives, limiting the formation of discriminative contrastive boundaries. Conversely, an excessively large value introduces difficult samples before stable representations are established, which may impair convergence.

The optimal values are $\lambda_0 = 0.05$ for *UCI* and *MOOC*, $\lambda_0 = 0.10$ for *Enron* and *UNtrade*, and $\lambda_0 = 0.20$ for *BITalpha*. In particular, *BITalpha* benefits from a relatively larger initial hard-negative ratio, whereas *UCI* and *MOOC* prefer a more conservative curriculum initialization. These results confirm that hard negatives should be introduced gradually with a moderate, dataset-specific initial difficulty.

Impact of RL Loss Weight α. Fig. 9 (right) presents the sensitivity to the RL loss weight α. A very small α provides insufficient optimization signals for the policy network, weakening the role of EDA in learning selective state transitions. In contrast, an excessively large α may cause the RL objective to interfere with the primary representation learning objective.

The optimal values are $\alpha = 0.15$, 0.10, 0.15, 0.20, and 0.05 for *UCI*, *Enron*, *MOOC*, *BITalpha*, and *UNtrade*, respectively. *BITalpha* favors stronger policy guidance, while *UNtrade* achieves its best result with a relatively small RL weight. Overall, the results demonstrate that the EFR-guided RL component should serve as an auxiliary regulatory mechanism rather than dominate the optimization process, and its contribution should be adjusted to the dynamic characteristics of each dataset.

5.6. Computational efficiency and scalability

Computational Complexity. Let N, E, T, M, K, and d denote the numbers of nodes, edges per snapshot, snapshots, evolutionary views, diffusion steps, and embedding dimensions, respectively. The overall time complexity is $\mathcal{O}!\left(MT((K+1)Ed + Nd^2)\right)$, including multi-view encoding, relational diffusion, and node-wise EDA. The space complexity is $\mathcal{O}!(M(Nd+E)+|\mathcal{B}|)$, where $|\mathcal{B}|$ denotes the historical-buffer size.

As shown in Table 9, RE-DGCL introduces moderate computational overhead. Under the same backbone architecture and training configuration, its average per-epoch training time is approximately 7.3% and 3.4% higher than Random Gate and Deterministic Gate, respectively, while the difference in peak GPU memory remains small. This compari-

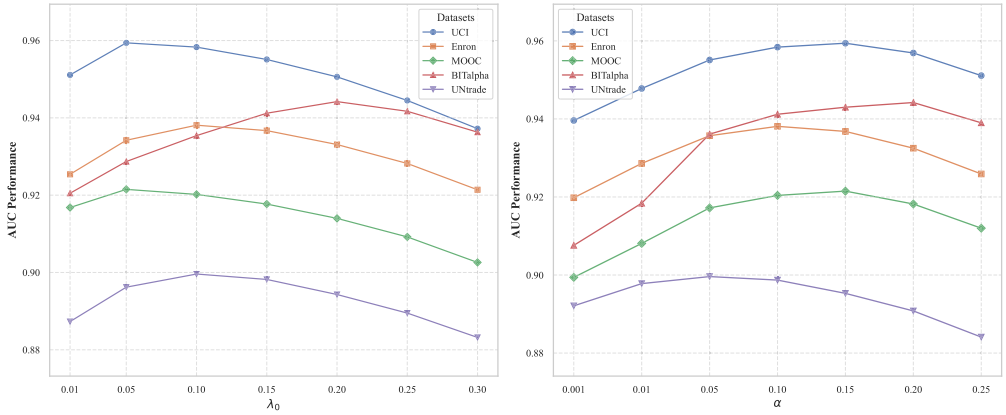


Fig. 9. Impact of λ_0 and α on model performance.

Table 9
Per-epoch training time (s) and peak GPU memory consumption (GB) on eight dynamic graph datasets.

(a) UCI, Enron, Reddit, and MOOC

Method	UCI		Enron		Reddit		MOOC	
	Time	Mem.	Time	Mem.	Time	Mem.	Time	Mem.
TGN	0.55	1.7	0.63	1.8	1.42	3.6	1.10	3.0
DyGFormer	0.82	2.2	0.91	2.4	2.08	4.8	1.61	4.0
CPDG	0.91	2.4	1.02	2.6	2.31	5.1	1.79	4.3
Random Gate	1.01	2.7	1.12	2.8	2.63	5.8	2.02	4.9
Deterministic Gate	1.04	2.7	1.15	2.9	2.71	5.9	2.07	5.0
RE-DGCL	1.08	2.8	1.20	3.0	2.78	6.0	2.14	5.1

(b) Fb-forum, BITalpha, UNtrade, and LastFM

Method	Fb-forum		BITalpha		UNtrade		LastFM	
	Time	Mem.	Time	Mem.	Time	Mem.	Time	Mem.
TGN	0.42	1.5	1.18	2.9	0.95	2.3	1.65	3.4
DyGFormer	0.68	2.0	1.76	3.8	1.32	3.1	2.42	4.7
CPDG	0.76	2.2	2.05	4.1	1.48	3.4	2.68	5.0
Random Gate	0.85	2.3	2.32	4.5	1.58	3.6	2.91	5.5
Deterministic Gate	0.91	2.4	2.36	4.9	1.68	3.6	3.03	5.7
RE-DGCL	0.94	2.5	2.46	5.0	1.72	3.7	3.12	5.8

Note: Time and Mem. denote per-epoch training time and peak GPU memory, respectively.

son suggests that the RL-driven decision mechanism incurs only limited marginal overhead relative to simpler gating strategies and does not alter the overall asymptotic complexity order. Overall, RE-DGCL maintains a practical accuracy–efficiency trade-off, although diffusion and historical-state storage may become bottlenecks on substantially larger graphs.

Overall, the experimental results provide complementary evidence for the main design choices of RE-DGCL. The comparison with representative dynamic graph baselines demonstrates its effectiveness across diverse interaction patterns, while the component ablations support the respective contributions of EDA, counterfactual EFR, and ENS. The controlled gating comparison further shows that the performance improvement is not solely attributable to reducing the update frequency, as the RL-driven policy performs better than forced, random, and deterministic alternatives under matched settings. In addition, the perturbation experiments demonstrate improved robustness to stochastic structural noise, and the efficiency analysis shows that the RL-driven decision mechanism introduces only limited marginal overhead compared with simpler gating strategies. Taken together, these results support the use of structurally guided selective evolution as an effective alternative to unconditional state updating in dynamic graph representation learning.

6. Conclusion and future work

This paper presents RE-DGCL, a Reinforcement-Evolutionary Dynamic Graph Contrastive Learning framework that transforms passive aggregation into proactive decisional evolution. RE-DGCL employs a bifurcated SGE–ALE architecture to separate stable structural information from local temporal dynamics. The RL-driven Evolutionary Decision Action selectively determines whether each interaction should trigger a state update, while the counterfactual Evolutionary Fidelity Reward evaluates update and maintenance candidates under the same structural-fidelity criterion. Meanwhile, curriculum-based Evolutionary-aware Negative Sampling retrieves temporally informative Hard Negative Constraints from historical interactions. Experiments on eight datasets and random-edge perturbation tests demonstrate consistent improvements in dynamic link prediction and robustness to stochastic structural noise.

RE-DGCL also has several limitations. When most interactions are regular and informative, selective evolution may behave similarly to forced updating, leading to smaller performance gains. In extremely sparse graphs, limited structural evidence may weaken the SGE anchor and policy feedback. Moreover, multi-view diffusion and historical buffers introduce additional computation and storage, while snapshot-conditional HNCs may include nodes that reconnect in later snapshots.

Several directions remain for future research. (1) **Scalability and Efficiency Optimization:** Scalable diffusion, adaptive subgraph sampling, and compressed historical buffers will be explored for large dynamic graphs. (2) **Generalization to Heterogeneous and Higher-order Dynamics:** The decisional paradigm will be extended to heterogeneous networks and higher-order interactions. (3) **Integration with Dynamic Graph Foundation Models:** Evolutionary sensing and decisional gating will be incorporated into general pre-training frameworks to improve transferability. (4) **False-negative-aware Evolutionary Sampling:** Uncertainty-aware HNC weighting and reconnection estimation will be investigated to reduce false-negative interference.

CRediT authorship contribution statement

Weidong Zhao: Resources, Funding acquisition; **Jingjing Liang:** Writing – original draft, Visualization, Software, Methodology; **Liqing Qiu:** Writing – review & editing, Supervision, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgement

This work was supported by the National Natural Science Foundation of China [Grant No. 72572095] and the Natural Science Foundation of Shandong Province of China [Grant No. ZR2023MF070].

References

Alemany, J., Del Val, E., & García-Fornes, A. (2023). A review of privacy decision-making mechanisms in online social networks. *ACM Computing Surveys*, *55*(2), 1–32.

Bei, Y., Xu, H., Zhou, S., Chi, H., Wang, H., Zhang, M., Li, Z., & Bu, J. (2023). Cpdg: A contrastive pre-training method for dynamic graph neural networks. *arXiv preprint*.

Bengio, Y., Louradour, J., Collobert, R., & Weston, J. (2009). Curriculum learning. *Proceedings of the 26th Annual International Conference on Machine Learning*, (pp. 41–48).

Bose, A. J., Ling, H., & Cao, Y. (2018). Adversarial contrastive estimation. *Proceedings of the 56th Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, (pp. 1021–1032).

Chen, T., Kornblith, S., Norouzi, M., & Hinton, G. (2020). A simple framework for contrastive learning of visual representations. In *International conference on machine learning* (pp. 1597–1607).

Cong, W., Zhang, S., Kang, J., Yuan, B., Wu, H., Zhou, X., Tong, H., & Mahdavi, M. (2023). Do we really need complicated model architectures for temporal networks? arXiv:2302.11636.

Fawcett, T. (2006). An introduction to roc analysis. *Pattern Recognition Letters*, *27*, 861–874.

Gao, K., Liu, C., Wu, J., Du, B., & Hu, W. (2024). Towards a better negative sampling strategy for dynamic graphs. *Neural Networks*, *173*, 106175.

Gao, Y., Yang, H., Zhang, P., Zhou, C., & Hu, Y. (2020). Graph neural architecture search. *Proceedings of the Twenty-Ninth International Joint Conference on Artificial Intelligence*, (pp. 1403–1409).

Goyal, P., Kamra, N., He, X., & Liu, Y. (2018). DynGEM: Deep embedding method for dynamic graphs. arXiv:1805.11273.

Grill, J.-B., Strub, F., Altché, F., Tallec, C., Richemond, P. H. et al. (2020). Bootstrap your own latent: A new approach to self-supervised learning. In *Advances in neural information processing systems* (pp. 21271–21284). (*volume. 33*).

Grover, A., & Leskovec, J. (2016). node2vec: Scalable feature learning for networks. *Proceedings of the 22nd ACM SIGKDD International Conference on Knowledge Discovery and Data Mining*, (pp. 855–864).

Guo, X., Yu, Z., Huang, F., Chen, X., Yang, D., & Wang, J. (2025). Dynamic meta graph convolutional recurrent network for heterogeneous spatiotemporal graph forecasting. *Neural Networks*, *181*, 106805.

Hamilton, W. L., Ying, R., & Leskovec, J. (2017). Inductive representation learning on large graphs. *Advances in Neural Information Processing Systems*, *30*.

Hassani, K., & Khasahmadi, A. H. (2020). Contrastive multi-view representation learning on graphs. In *Proceedings of the 37th international conference on machine learning* (pp. 4116–4126).

He, D., Zhao, J., Huo, C., Huang, Y., Huang, Y., & Feng, Z. (2024). A new mechanism for eliminating implicit conflict in graph contrastive learning. *Proceedings of the AAAI Conference on Artificial Intelligence*, *38*(11), 12340–12348.

He, K., Fan, H., Wu, Y., Xie, S., & Girshick, R. (2020). Momentum contrast for unsupervised visual representation learning. In *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition* (pp. 9726–9735).

Huang, S., Poursafaei, F., Danovitch, J., Fey, M., Hu, W., Rossi, E. et al. (2023). Temporal graph benchmark for machine learning on temporal graphs. arXiv:2307.01026.

Huang, Y., Zhao, J., He, D., Jin, D., Huang, Y., & Wang, Z. (2025). Does GCL need a large number of negative samples? Enhancing graph contrastive learning with effective and efficient negative sampling. *Proceedings of the AAAI Conference on Artificial Intelligence*, *39*(16), 17511–17518.

Kaelbling, L. P., Littman, M. L., & Moore, A. W. (1996). Reinforcement learning: A survey. *Journal of Artificial Intelligence Research*, *4*, 237–285.

Kazemi, S. M., Goel, R., Jain, K., Kobyzev, I., Sethi, A., Forsyth, P., & Poupart, P. (2020). Representation learning for dynamic graphs: A survey. *Journal of Machine Learning Research*, *21*, 1–73.

Kipf, T. N., & Welling, M. (2016). Semi-supervised classification with graph convolutional networks. arXiv:1609.02907.

Kumar, S., Spezzano, F., Subrahmanian, V., & Faloutsos, C. (2016). Edge weight prediction in weighted signed networks. In *Proceedings of the 22nd ACM SIGKDD international conference on knowledge discovery and data mining* (pp. 242–251).

Kumar, S., Zhang, X., & Leskovec, J. (2019a). Predicting dynamic embedding trajectory in temporal interaction networks. In *Proceedings of the 25th ACM SIGKDD international conference on knowledge discovery & data mining* (pp. 1269–1278).

Kumar, S., Zhang, X., & Leskovec, J. (2019b). Predicting dynamic embedding trajectory in temporal interaction networks. In *Proceedings of the 25th ACM SIGKDD international conference on knowledge discovery and data mining* (pp. 1269–1278).

Kumaran, G., & Allan, J. (2008). Adapting information retrieval systems to user queries. *Information Processing and Management*, *44*, 1838–1862.

Li, D., Kosugi, S., Zhang, Y., Okumura, M., Xia, F., & Jiang, R. (2025a). Revisiting dynamic graph clustering via matrix factorization. In *Proceedings of the ACM on web conference 2025* (pp. 1342–1352). Association for Computing Machinery.

Li, D., Tan, S., Zhang, Y., Jin, M., Pan, S., Okumura, M., & Jiang, R. (2025b). Dyg-mamba: Continuous state space modeling on dynamic graphs. In *Advances in neural information processing systems* (pp. 129101–129130). Curran Associates, Inc. (*volume. 38*).

Li, H., Li, C., Feng, K., Yuan, Y., Wang, G., & Zha, H. (2024). Robust knowledge adaptation for dynamic graph neural networks. *IEEE Transactions on Knowledge and Data Engineering*, *36*, 6920–6933.

Li, J., Zhou, P., Xiong, C., & Hoi, S. C. H. (2021). Prototypical contrastive learning of unsupervised representations. *ICLR*.

Li, Y., Tarlow, D., Brockschmidt, M., & Zemel, R. (2016). Gated graph sequence neural networks. *ICLR*.

Ma, Y., Guo, Z., Ren, Z., Tang, J., & Yin, D. (2020). Streaming graph neural networks. *Proceedings of the 43rd International ACM SIGIR Conference on Research and Development in Information Retrieval*, (pp. 719–728).

MacDonald, G. K., Brauman, K. A., Sun, S., Carlson, K. M. et al. (2015). Rethinking agricultural trade relationships in an era of globalization. *Proceedings of the National Academy of Sciences*, *112*(17), 5309–5314.

Mikolov, T., Chen, K., Corrado, G., & Dean, J. (2013). Efficient estimation of word representations in vector space. *arXiv preprint*.

Nguyen, G. H., Lee, J. B., Rossi, R. A., Ahmed, N. K., Koh, E., & Kim, S. (2018). Continuous-time dynamic network embeddings. In *Companion of the the web conference 2018* (pp. 969–976).

Opsahl, T., & Panzarasa, P. (2009). *Clustering in weighted networks*. (pp. 155–163). Elsevier (*volume. 31*).

Panzarasa, P., Opsahl, T., & Carley, K. M. (2009). Patterns and dynamics of users' behavior and interaction: Network analysis of an online community. (pp. 155–163). (*volume. 31*).

Pareja, A., Domeniconi, G., Chen, J., Ma, T., Suzumura, T., Kanezashi, H., Kaler, T., Schardl, T. B., & Leiserson, C. E. (2020). EvolveGCN: Evolving graph convolutional networks for dynamic graphs. In *Proceedings of the AAAI conference on artificial intelligence* (pp. 5363–5370).

Perozzi, B., Al-Rfou, R., & Skiena, S. (2014). Deepwalk: Online learning of social representations. In *Proceedings of the 20th ACM SIGKDD international conference on knowledge discovery and data mining* (pp. 701–710).

Qian, L., Zuo, Q., Li, D., & Zhu, H. (2025). Dgmscl: A dynamic graph mixed supervised contrastive learning approach for class imbalanced multivariate time series classification. *Neural Networks*, *185*, 107131.

Qin, Y., Ju, W., Gu, Y., Qiao, Z., Xiao, Z., & Zhang, M. (2025). PolyCF: Towards optimal spectral graph filters for collaborative filtering. *ACM Transactions on Information Systems*, *43*(4), 1–28.

Rossi, E., Chamberlain, B., Frasca, F., Eynard, D., Monti, F., & Bronstein, M. M. (2020). Temporal graph networks for deep learning on dynamic graphs. arXiv:2006.10637.

Sankar, A., Wu, Y., Gou, L., Zhang, W., & Yang, H. (2020). DySAT: Deep neural representation learning on dynamic graphs via self-attention networks. In *Proceedings of the 13th ACM international conference on web search and data mining* (pp. 519–527). ACM.

Seo, Y., Defferrard, M., Vandergheynst, P., & Bresson, X. (2018). Structured sequence modeling with graph convolutional recurrent networks. In *International conference on learning representations*.

Sharma, K., Lee, Y.-C., Nambi, S., Salian, A., Shah, S., Kim, S.-W., & Kumar, S. (2024). A survey of graph neural networks for social recommender systems. *ACM Computing Surveys*, *56*(10), 1–34.

Shetty, J., & Adibi, J. (2004). The enron email dataset database schema and brief statistical report.

Skarding, J., Gabrys, B., & Musial, K. (2021). Foundations and modelling of dynamic networks using dynamic graph neural networks: A survey. *IEEE Access*, *9*, 79143–79168.

Tan, S., Li, D., Jiang, R., Zhang, Y., & Okumura, M. (2024). Community-invariant graph contrastive learning. In *Proceedings of the 41st international conference on machine learning* (pp. 47579–47606). PMLR (*volume. 235*).

Tang, J., Qu, M., Wang, M., Zhang, M., Yan, J., & Mei, Q. (2015). Line: Large-scale information network embedding. In *Proceedings of the 24th international conference on world wide web* (pp. 1067–1077).

Thakoor, S., Tallec, C., Azar, M. G., Azabou, M., Dyer, E. L., Munos, R., Veličković, P., & Valko, M. (2022). Large-scale representation learning on graphs via bootstrapping. In *International conference on learning representations*.

Trivedi, R., Farajtabar, M., Biswal, P., & Zha, H. (2019). Dyrep: Learning representations over dynamic graphs. *International Conference on Learning Representations*.

Veličković, P., Fedus, W., Hamilton, W. L., Liò, P., Bengio, Y., & Hjelm, R. D. (2018). Deep graph infomax. arXiv:1809.10341.

Wei, X., Gong, X., Zhan, Y., Du, B., Luo, Y., & Hu, W. (2023). Clnode: Curriculum learning for node classification. *Proceedings of the Sixteenth ACM International Conference on Web Search and Data Mining*, (pp. 670–678).

Xia, J., Wu, L., Wang, G., Chen, J., & Li, S. Z. (2022). ProGCL: Rethinking hard negative mining in graph contrastive learning. In *Proceedings of the 39th international conference on machine learning* (pp. 24332–24346). PMLR (*volume. 162*).

Xu, D., Ruan, C., Korpeoglu, E., Kumar, S., & Achan, K. (2020). Inductive representation learning on temporal graphs. *ICLR*.

Xu, L., Pan, Z., Chen, H., & Wang, S. (2024a). Mtdyg: Multi-scale transformers with continuous time dynamic graph model for link prediction. *Knowledge-Based Systems*, *300*, 112245.

Xu, Y., Peng, Z., Shi, B., Hua, X., & Dong, B. (2024b). Learning dynamic graph representations through timespan view contrasts. *Neural Networks*, *176*, 106384.

Yi, S., Ju, W., Qin, Y., Luo, X., Liu, L., Zhou, Y., & Zhang, M. (2024). Redundancy-free self-supervised relational learning for graph clustering. *IEEE Transactions on Neural Networks and Learning Systems*, *35*(12), 18313–18327.

Ying, R., He, R., Chen, K., Eksombatchai, P., Hamilton, W. L., & Leskovec, J. (2018). Graph convolutional neural networks for web-scale recommender systems. *Proceedings of the 24th ACM SIGKDD International Conference on Knowledge Discovery and Data Mining*, (pp. 974–983).

You, J., Ying, R., & Leskovec, J. (2019). Position-aware graph neural networks. *Proceedings of the 36th International Conference on Machine Learning*, (pp. 7134–7143).

Yu, B., Yin, H., & Zhu, Z. (2018). Spatio-temporal graph convolutional networks: A deep learning framework for traffic forecasting. In *Proceedings of the 27th international joint conference on artificial intelligence* (pp. 3634–3640).

Yu, L., Sun, L., Du, B., & Lv, W. (2023). Towards better dynamic graph learning: New architecture and unified library. arXiv:2303.13047.

Zeng, X., Peng, H., & Li, A. (2026). Proactive bot detection based on structural information principles. *IEEE Transactions on Pattern Analysis and Machine Intelligence*, *48*(4), 4726–4743.

Zeng, X., Peng, H., Su, D., & Li, A. (2025). Hierarchical decision making based on structural information principles. *Journal of Machine Learning Research*, *26*(182), 1–55.

Zhang, W., Chen, T., Wang, J., & Yu, Y. (2013). Optimizing top-n collaborative filtering via dynamic negative item sampling. *Proceedings of the 36th International ACM SIGIR Conference on Research and Development in Information Retrieval*, (pp. 785–788).

Zhu, Y., Xu, Y., Yu, F., Liu, Q., Wu, S., & Wang, L. (2020). Deep graph contrastive representation learning. arXiv:2006.04131.

Zhu, Y., Xu, Y., Yu, F., Liu, Q., Wu, S., & Wang, L. (2021). Graph contrastive learning with adaptive augmentation. In *Proceedings of the web conference 2021* (pp. 2069–2080).